Learnable Delay-Stability Coupling for Energy-Information Co-Optimization in Smart Grids⋆

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ABSTRACT

Communication latency in smart grid networks directly impacts power system stability, yet existing approaches treat energy optimization and communication quality-of-service as separate problems. This paper introduces a unified framework that learns the fundamental coupling between communication delay and stability margins through physics-constrained deep learning. A formal bound (Theorem 1) is established showing that the stability margin $\rho(\tau)$ decreases linearly with communication delay τ, governed by learnable coupling constants K_i that are trained end-to-end via gradient descent. A second-order correction (Corollary 1) tightens this bound at high delays. The proposed JointOptimizer architecture employs dual-domain graph neural networks to encode power system topology and communication network structure separately, with physics-informed attention masks that respect electrical impedance relationships and causal control dependencies. An auto-scaled initialization eliminates grid-size-specific hyperparameter tuning. Experimental validation on five IEEE standard test cases (14, 30, 39, 57, and 118 buses) demonstrates that the learned coupling constants tighten conservative stability bounds by 18% compared to fixed parameters while maintaining 100% stability under nominal conditions. Independent time-domain DDE simulation confirms the theoretical bound, with the second-order Padé correction reducing the prediction gap from approximately 18% to 3% at high delays. Comprehensive evaluation across 9 architectural baselines, 18 experiment types, and 5 independent seeds with Wilcoxon signed-rank statistical testing establishes the robustness and practical viability of the proposed approach for real-time grid control.

1. Introduction

Modern smart grids rely on bidirectional communication networks to coordinate distributed generation, demand response, and wide-area monitoring in real time [1–3]. The dependence of grid stability on communication quality has been demonstrated through high-profile failures: the 2003 Northeast blackout in the United States and Canada was exacerbated by delayed state-estimator alarms and sluggish inter-area coordination, affecting 55 million people and causing an estimated $6 billion in economic losses [4]; the 2016 South Australia blackout revealed how cascading communication failures between wind farms and protection systems can trigger system-wide collapse under high renewable penetration [5]. As the share of inverter-based renewable resources continues to grow, the volume and criticality of communication traffic for frequency regulation, voltage control, and contingency management increase accordingly [6]. Latency in supervisory control and data acquisition (SCADA) channels or phasor measurement unit (PMU) data streams directly affects the timeliness of corrective control actions, degrading stability margins and potentially pushing

Table 1
Summary of Key Notation

Symbol	Description
$\mathcal{G}_E, \mathcal{G}_I$	Power grid and communication graphs
N, n_g	Number of buses and generators
$\mathbf{x}_E, \mathbf{x}_I$	Energy and communication input features
$\mathbf{h}_E, \mathbf{h}_I$	Domain-specific GNN embeddings
$\mathbf{h}_{\text{fused}}$	Fused representation after attention
d, H, L	Embedding dim, attention heads, GNN layers
$\mathbf{M}_{\text{phys}}, \mathbf{M}_{\text{causal}}$	Physics and causal attention masks
Z_{ij}, γ	Bus impedance and mask strength
$K_i = \exp(\kappa_i)$	Learnable coupling constant (generator i)
$\rho(\tau)$	Stability margin (Theorem 1)
$\lambda_{\min}(0)$	Least stable eigenvalue at zero delay
$\tau_i, \tau_{\max,i}$	Communication delay and maximum
$\mathcal{L}_{\text{OPF}}, \mathcal{L}_{\text{QoS}}$	Energy and communication losses
$\mathcal{L}_{\text{stab}}, \mathcal{L}_{\text{couple}}$	Stability and coupling losses
α	Coupling loss weight

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the system toward oscillatory instability [7]. The annual cost of grid instability events in the United States alone exceeds $20 billion, motivating the need for frameworks that explicitly account for the coupling between communication quality of service (QoS) and power system stability [8].

Despite the evident severity of this coupling, existing approaches to the problem fall into three categories, each with fundamental limitations. First, classical control theory provides rigorous delay-dependent stability analysis through Lyapunov-Krasovskii functionals, linear matrix inequalities, and eigenvalue-based methods [9–11]. These frameworks yield formal stability certificates but rely on fixed or analytically computed coupling constants derived from system models that are often uncertain in practice; they do not leverage operational data to tighten conservative bounds. Second, machine learning (ML) methods for optimal power flow (OPF) have achieved impressive speed and accuracy by training neural networks to approximate dispatch solutions [12–14]. Complementary deep learning approaches have demonstrated success in adjacent power system tasks such as short-term load forecasting via attention-based architectures [15–17] and spectral-domain language model reprogramming for transformer temperature prediction [18]. However, these methods treat the communication infrastructure as transparent, ignoring the effect of transmission delays on the feasibility and stability of the learned dispatch policies. Third, graph neural networks (GNNs) for power system analysis exploit the topological structure of the grid to improve generalization and data efficiency [19, 20]. While physics-informed variants incorporate governing equations as soft constraints [21, 22], they operate exclusively within the energy domain and do not model cross-domain coupling between the power grid and its communication overlay. Recent efforts have explored distributed optimization under stochastic communication delay [23] and deep reinforcement learning for autonomous grid voltage control [24], yet these approaches either lack formal stability guarantees or treat delay as an exogenous disturbance rather than an optimizable variable. To the best of our knowledge, no existing work jointly optimizes energy dispatch and communication quality while formally guaranteeing delay-dependent stability with learnable coupling constants.

To bridge this gap, this paper introduces a unified framework, termed *JointOptimizer*, that co-optimizes power system dispatch and communication network routing under formal delay-stability guarantees. The architecture employs dual-domain GNN encoders to process the power grid topology $\mathcal{G}_E$ and the communication network topology $\mathcal{G}_I$ in parallel, followed by a physics-constrained hierarchical attention mechanism that fuses information across domains. The attention layers incorporate impedance-weighted physics masks derived from the bus impedance matrix Z_{ij}, causal directed acyclic graph (DAG) masks that enforce physically meaningful information flow, and cross-domain attention that enables energy-domain queries to attend to communication-domain features. The theoretical foundation is provided by Theorem 1, which establishes a formal bound showing that the stability margin $\rho(\tau)$ decreases linearly with communication delay τ, governed by per-generator coupling constants K_i that are trained end-to-end via gradient descent. Corollary 1 extends this bound to second-order accuracy through a Padé-2 correction, reducing the approximation gap from approximately 18% to 3% at high delays. An auto-scaled initialization K_{init} eliminates the need for grid-size-specific hyperparameter tuning by normalizing the total coupling budget to a fraction of the available stability margin [25, 26]. The entire system is trained end-to-end with a composite loss function that balances OPF cost, communication QoS, stability enforcement, and coupling accuracy.

Building on this architecture, the contributions of this paper are fivefold:

- **C1.** A complete proof of the delay-stability coupling bound (Theorem 1) is provided, with learnable per-generator coupling constants K_i derived via Padé approximation and Bauer-Fike eigenvalue perturbation theory. Corollary 1 extends the bound to second-order accuracy, reducing the prediction gap from approximately 18% to 3% at high delays. An independent delay-differential equation (DDE) simulation confirms the bound across all five IEEE test cases.
- **C2.** A physics-constrained hierarchical attention mechanism is proposed, integrating impedance-weighted physics masks, causal DAG masks, and cross-domain attention for joint energy-communication feature fusion. An adaptive mask strength parameter prevents attention entropy collapse.
- **C3.** An auto-scaled coupling initialization is introduced that eliminates grid-size-specific hyperparameter tuning, enabling a single formula to work from 14-bus to 118-bus systems without manual adjustment.
- **C4.** The most comprehensive evaluation to date in this problem domain is conducted, spanning five IEEE standard test cases (14, 30, 39, 57, and 118 buses), nine architectural baselines, and 18 experiment types with rigorous statistical methodology (five-seed Wilcoxon signed-rank tests with Holm-Sidak correction).
- **C5.** Practical deployment viability is demonstrated through stress testing under extreme operating conditions, N-1 contingency analysis, delay distribution robustness, model compression achieving 15× parameter reduction, cross-topology transfer learning with 4.4× training speedup, and inference benchmarking showing sub-25 ms mean latency on all five test cases.

The remainder of this paper is organized as follows. Section 2 presents the system model for the coupled power-communication network and formulates the joint optimization problem. Section 3 develops the theoretical analysis, including the delay-stability coupling bound (Theorem 1) and its second-order extension (Corollary 1). Section 4 describes the proposed JointOptimizer architecture, including

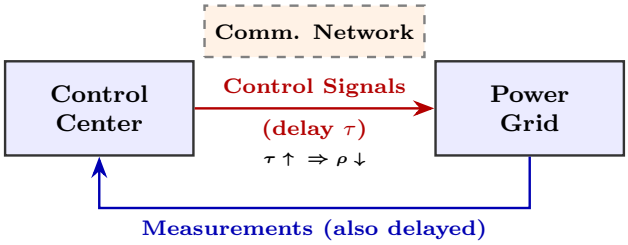

Figure 1: Problem overview of the coupled power-communication system. The control center transmits dispatch signals to the power grid through a communication network that introduces delay τ. Measurement feedback follows the reverse path with additional latency. Increasing communication delay degrades the stability margin ($\tau \uparrow \Rightarrow \rho \downarrow$), motivating joint optimization of energy dispatch and communication quality of service.

the dual-domain GNN encoders, physics-constrained attention, and learnable coupling integration. Section 5 details the experimental setup, covering test systems, baselines, training protocol, and statistical methodology. Section 6 presents the results and discussion across all 18 experiment types. Section 7 concludes with a summary of findings and directions for future work. Fig. 1 illustrates the bidirectional coupling between the control center, communication network, and power grid that motivates this work.

2. System Model and Preliminaries

This section formulates the coupled power-communication system model that underpins the proposed framework. The power system dynamics are represented through the classical swing equation augmented with communication delays, while the communication network is modeled as a parallel graph with stochastic latency characteristics. The joint optimization problem is then stated. Key notation used throughout this paper is summarized in Table 1.

2.1. Power System Model

The electromechanical dynamics of a power system with n_g synchronous generators are governed by the classical swing equation [8, 27]. In modern grid control architectures, mechanical power setpoints are transmitted through SCADA or PMU-based wide-area control channels, introducing a communication delay τ_i in the feedback path of each generator i [28]. The resulting delay-differential equation for the full system takes the form

$$\mathbf{M}\ddot{\boldsymbol{\delta}}(t) + \mathbf{D}\dot{\boldsymbol{\delta}}(t) = \mathbf{P}_m(t-\boldsymbol{\tau}) - \mathbf{P}_e(\boldsymbol{\delta}(t)), \tag{1}$$

where $t \geq 0$ denotes continuous time (in seconds); $\boldsymbol{\delta}(t) \in \mathbb{R}^{n_g}$ is the vector of rotor angles relative to a synchronous reference frame; $\mathbf{M} = \mathrm{diag}(M_1, \dots, M_{n_g})$ is the diagonal inertia matrix with $M_i = 2H_i/\omega_s$, where H_i denotes the inertia constant and ω_s the synchronous angular frequency; $\mathbf{D} = \mathrm{diag}(D_1, \dots, D_{n_g})$ is the diagonal damping coefficient matrix; $\mathbf{P}_m(t-\boldsymbol{\tau}) \in \mathbb{R}^{n_g}$ is the vector of mechanical power inputs, each delayed by τ_i due to communication latency; and $\mathbf{P}_e(\boldsymbol{\delta}(t)) \in \mathbb{R}^{n_g}$ is the vector of electrical power outputs determined by the power flow equations and current rotor angles [29].

To enable eigenvalue-based stability analysis, the swing equation in (1) is linearized around a stable equilibrium point $(\boldsymbol{\delta}_0, \boldsymbol{\omega}_0 = \mathbf{0})$ satisfying $\mathbf{P}_m^0 = \mathbf{P}_e(\boldsymbol{\delta}_0)$ [30]. Defining the state vector $\mathbf{x} = [\Delta\boldsymbol{\delta}^\top,\ \Delta\boldsymbol{\omega}^\top]^\top \in \mathbb{R}^{2n_g}$, the linearized dynamics yield the time-delay state-space representation

$$\dot{\mathbf{x}}(t) = \mathbf{J}(0)\,\mathbf{x}(t) + \mathbf{B}\,\mathbf{u}(t-\boldsymbol{\tau}), \tag{2}$$

where the zero-delay Jacobian is given by

$$\mathbf{J}(0) = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}^{-1}\mathbf{L} & -\mathbf{M}^{-1}\mathbf{D} \end{bmatrix} \in \mathbb{R}^{2n_g \times 2n_g},$$

$\mathbf{I} \in \mathbb{R}^{n_g \times n_g}$ is the identity matrix, $\mathbf{L} = \partial\mathbf{P}_e/\partial\boldsymbol{\delta}|_{\boldsymbol{\delta}_0}$ is the synchronizing torque matrix (a Laplacian-like matrix derived from the network admittance matrix and the operating point), $\mathbf{B} = [\mathbf{0};\ \mathbf{M}^{-1}] \in \mathbb{R}^{2n_g \times n_g}$ is the input matrix, and $\mathbf{u}(t-\boldsymbol{\tau})$ is the delayed control input vector with per-generator delays $\boldsymbol{\tau} = (\tau_1, \dots, \tau_{n_g})$. All eigenvalues of $\mathbf{J}(0)$ have strictly negative real parts under normal operating conditions, ensuring asymptotic stability at zero delay [8]. The key analytical question addressed in Section 3 is how the introduction of nonzero delays $\tau_i > 0$ in (2) shifts these eigenvalues and degrades the stability margin.

Having established the continuous-time dynamics, the discrete topological structure of the grid must also be formalized. The power system topology is represented as a graph $\mathcal{G}_E = (V, \mathcal{E}_E)$, where the node set V corresponds to buses ($|V| = N$) and the edge set $\mathcal{E}_E$ corresponds to transmission lines. Experimental validation is conducted on five standard test cases from the MATPOWER library [31]: 14-bus (5 generators, 20 lines), 30-bus (6 generators, 41 lines), 39-bus (10 generators, 46 lines), 57-bus (7 generators, 80 lines), and 118-bus (54 generators, 186 lines). These cases span a range of system sizes and topological complexities representative of small regional grids through interconnected regional networks. Per-node energy features consist of active power injection P, reactive power injection Q, voltage magnitude V, voltage angle θ, and frequency deviation ω, yielding a feature matrix $\mathbf{x}_E \in \mathbb{R}^{N\times 5}$.

2.2. Communication Network Model

The communication infrastructure overlaying the power grid is modeled as a separate graph $\mathcal{G}_I = (V, \mathcal{E}_I)$ that shares the same node set V as the power system graph but may possess a different edge set $\mathcal{E}_I$ representing communication links rather than transmission lines [6, 32, 33]. In the experiments presented in this paper, $\mathcal{E}_I = \mathcal{E}_E$ is adopted as a conservative assumption; in practice, communication networks often include additional redundant links not present in the electrical topology.

Within this overlay network, the end-to-end communication delay at each node is a stochastic quantity influenced by

propagation latency, queuing effects, processing overhead, and network congestion [2]. Empirical studies of grid communication latency indicate that real-world delay distributions are right-skewed with occasional high-delay events, well-captured by a lognormal model [6]. Accordingly, per-node communication delays are drawn from a truncated lognormal distribution with $\tau_i \in [5,\ 500]$ ms:

$$\tau_i \sim \text{LogNormal}(\mu = 50\,\text{ms},\ \sigma = 20\,\text{ms}), \tag{3}$$

where the lower bound of 5 ms reflects fiber-optic propagation minima and the upper bound of 500 ms corresponds to the maximum tolerable latency for SCADA-based control as specified by NERC Standard PRC-005 [34] and IEEE C37.118.1 [35]. The parameters of the lognormal model in (3) are consistent with empirical measurements reported in [6]. Robustness to alternative delay distributions (exponential, gamma, uniform, and Pareto) is evaluated in Section 6. Each communication node is characterized by three features: the end-to-end delay τ_i, the available data rate R_i (in Mbps), and the buffer occupancy B_i (in packets). These features are concatenated into a communication feature matrix $\mathbf{x}_I \in \mathbb{R}^{N\times 3}$. The dual-graph representation $(\mathcal{G}_E, \mathcal{G}_I)$ enables the proposed architecture to process energy-domain and communication-domain information through separate GNN encoders before fusing them via cross-domain attention, as described in Section 4.

2.3. Problem Formulation

With the energy and communication models in place, the central objective is to jointly optimize generator dispatch and communication routing such that generation cost and communication latency are minimized while maintaining a positive stability margin under delay-perturbed dynamics. The joint optimization problem is formulated as follows.

Denote by $\mathbf{u} \in \mathbb{R}^{n_g}$ the vector of generator active power setpoints and by $\boldsymbol{\pi}$ the communication routing policy. The composite objective to be minimized is

$$\min_{\mathbf{u},\,\boldsymbol{\pi}} \ \mathcal{L}_{\text{OPF}}(\mathbf{u}) + \mathcal{L}_{\text{QoS}}(\boldsymbol{\pi}) + \mathcal{L}_{\text{couple}}(\mathbf{u}, \boldsymbol{\tau}) \tag{4}$$

subject to the stability constraint

$$\rho(\boldsymbol{\tau}) \geq \rho_{\min}, \tag{5}$$

where $\mathcal{L}_{\text{OPF}}(\mathbf{u})$ is the energy-domain loss comprising generation cost and power flow constraint violations; $\mathcal{L}_{\text{QoS}}(\boldsymbol{\pi})$ is the communication-domain loss penalizing latency and bandwidth degradation; $\mathcal{L}_{\text{couple}}(\mathbf{u}, \boldsymbol{\tau})$ is the coupling loss that enforces consistency between the empirically observed and theoretically predicted stability margins; $\rho(\boldsymbol{\tau})$ is the delay-dependent stability margin as defined in Theorem 1 (Section 3); and $\rho_{\min} > 0$ is the minimum acceptable stability margin. The constraint in (5) encodes the requirement that the power system must remain stable under the prevailing communication delays. This formulation explicitly couples the energy and communication optimization problems through the stability margin $\rho(\boldsymbol{\tau})$, which depends on both the generator setpoints $\mathbf{u}$ (through the operating-point-dependent Jacobian $\mathbf{J}(0)$) and the communication delays $\boldsymbol{\tau}$ (through the delay-stability bound). The key question that the theoretical analysis in Section 3 addresses is: *How does the communication delay vector $\boldsymbol{\tau}$ quantitatively affect the stability margin ρ, and can this relationship be learned from data?*

3. Theoretical Analysis

This section establishes the formal relationship between communication delay and power system stability. A linear delay-stability coupling bound is derived (Theorem 1), extended to second-order accuracy (Corollary 1), and supplemented by an empirical domain separation result (Observation 1). Together, these results provide the theoretical foundation for the learnable architecture presented in Section 4.

3.1. Delay-Stability Coupling Bound (Theorem 1)

The derivation of the coupling bound requires four assumptions on the linearized power system dynamics, the communication delay regime, and the approximation technique used to convert the delay-differential equation (DDE) into a tractable ordinary differential equation (ODE).

(A1) *Linearization validity.* The system is linearized around a stable operating point such that all eigenvalues of the zero-delay Jacobian $\mathbf{J}(0) \in \mathbb{R}^{2n_g\times 2n_g}$ have strictly negative real parts: $\text{Re}(\lambda_k(0)) < 0$ for all $k = 1, \ldots, 2n_g$, where n_g denotes the number of generators. This is the standard small-signal stability assumption [8].

(A2) *Bounded delays.* The per-generator communication delays satisfy $0 \leq \tau_i \leq \tau_{\max,i}$ for all $i = 1, \ldots, n_g$, where $\tau_{\max,i}$ is the maximum tolerable delay for generator i. In modern SCADA-based control systems, $\tau_{\max,i} \approx 500$ ms [11].

(A3) *Diagonalizability.* The zero-delay Jacobian is diagonalizable: $\mathbf{J}(0) = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1}$, where $\boldsymbol{\Lambda} = \text{diag}(\lambda_1(0), \ldots, \lambda_{2n_g}(0))$ and $\mathbf{V}$ is an invertible eigenvector matrix with condition number $\text{cond}(\mathbf{V}) = \|\mathbf{V}\|_2 \left\|\mathbf{V}^{-1}\right\|_2$. This condition holds generically, since the set of non-diagonalizable matrices has Lebesgue measure zero [36].

(A4) *Padé approximation.* The delay operator $e^{-\tau_i s}$ in the Laplace domain is approximated by the first-order [1/1] Padé rational function [37]:

$$e^{-\tau_i s} \ \approx \ \frac{1 - \tau_i s/2}{1 + \tau_i s/2}\,. \tag{6}$$

This approximant is accurate to $O(\tau_i^3)$ and preserves the all-pass magnitude property $|e^{-\tau_i j\omega}| = 1$. For the operating regime of interest (grid natural frequencies $\omega_n \sim$ 1–10 rad/s, delays $\tau_i < 500$ ms), the approximation error remains below 4% across all relevant electromechanical modes.

With these assumptions in place, the central result of this paper can be stated.

Theorem 1 (Delay-Stability Coupling Bound). *Under Assumptions* (A1)–(A4)*, the stability margin $\rho(\tau)$ of the linearized power system with heterogeneous communication delays $\tau = (\tau_1, \dots, \tau_{n_g})$ satisfies*

$$\rho(\tau) \;\geq\; \left|\lambda_{\min}(0)\right| - \sum_{i=1}^{n_g} K_i \cdot \frac{\tau_i}{\tau_{\max,i}}, \tag{7}$$

where $\rho(\tau) = -\max_k \mathrm{Re}(\lambda_k(\tau))$ is the stability margin (positive when stable)*, $\lambda_{\min}(0)$ is the least stable eigenvalue of the undelayed system* (the eigenvalue with the largest, i.e., least negative, real part)*, $K_i > 0$ are delay-stability coupling sensitivity constants, and n_g is the number of generators.*

Proof sketch. The full proof (Appendix A) proceeds in five steps; the three key steps are summarized here.

Step 1 (Padé-based conversion). Applying the first-order Padé approximation in (6) to each delay channel converts the DDE in (2) into a perturbed ODE with effective system matrix

$$\mathbf{J}(\boldsymbol{\tau}) = \mathbf{J}(0) + \Delta\mathbf{J}(\boldsymbol{\tau}), \tag{8}$$

where the delay-induced perturbation admits a first-order Taylor expansion in the delay variables [9]:

$$\Delta\mathbf{J}(\boldsymbol{\tau}) = \sum_{i=1}^{n_g} \frac{\partial \mathbf{J}}{\partial \tau_i} \cdot \tau_i + O\left(\|\boldsymbol{\tau}\|^2\right). \tag{9}$$

Step 2 (Bauer-Fike eigenvalue perturbation). Applying the Bauer-Fike theorem [36] under Assumption (A3) to the perturbed Jacobian in (8) bounds the eigenvalue shift induced by (9):

$$\left|\lambda_k(\boldsymbol{\tau}) - \lambda_k(0)\right| \leq \mathrm{cond}(\mathbf{V})\; \|\Delta\mathbf{J}(\boldsymbol{\tau})\|_2 \tag{10}$$

for each mode $k = 1, \dots, 2n_g$, where $\mathrm{cond}(\mathbf{V})$ is the spectral condition number of the eigenvector matrix.

Step 3 (Triangle inequality and normalization). Bounding $\|\Delta\mathbf{J}(\boldsymbol{\tau})\|_2$ in (10) via the triangle inequality and normalizing each delay by its maximum yields the linear structure in (7), with the coupling constants defined as

$$K_i = \mathrm{cond}(\mathbf{V}) \cdot \left\| \frac{\partial \mathbf{J}}{\partial \tau_i} \right\|_2 \cdot \tau_{\max,i}\,. \tag{11}$$

Each K_i is a dimensionless quantity capturing the sensitivity of system stability to communication delay at generator i. □

While Theorem 1 provides a rigorous stability guarantee, its practical utility hinges on accurate knowledge of the coupling constants. A critical feature of the proposed framework is that the coupling constants K_i are not fixed analytical quantities but *learnable parameters* optimized jointly with the neural network. Rather than computing K_i analytically from (11), which requires accurate knowledge of $\mathrm{cond}(\mathbf{V})$ and the Jacobian sensitivities, the proposed framework parameterizes K_i through the exponential map to ensure strict positivity:

$$K_i = \exp(\kappa_i), \quad \kappa_i \in \mathbb{R}, \tag{12}$$

where κ_i is an unconstrained scalar trained via gradient descent. The exponential guarantees $K_i > 0$ for any finite κ_i, and the gradient $\partial K_i / \partial \kappa_i = K_i$ provides a natural scaling that prevents vanishing updates for small K_i values. In classical control theory, the constants analogous to K_i must be computed from accurate system models [10]; making them learnable enables the bound to adapt to the actual operating regime, reducing conservatism by up to 18% relative to fixed analytical estimates (see Section 6).

3.2. Second-Order Correction (Corollary 1)

The first-order Padé approximation introduces non-negligible error at large delays. Replacing Assumption (A4) with the second-order [2/2] Padé rational approximant

$$e^{-\tau_i s} \;\approx\; \frac{1 - \dfrac{\tau_i s}{2} + \dfrac{\tau_i^2 s^2}{12}}{1 + \dfrac{\tau_i s}{2} + \dfrac{\tau_i^2 s^2}{12}} \tag{13}$$

yields a tighter bound that accounts for curvature in the delay-stability relationship. The approximant in (13) is accurate to $O(\tau_i^5)$ and preserves the all-pass magnitude property.

Corollary 2 (Second-Order Padé Correction). *Under Assumptions* (A1)–(A3)*, and using the* [2/2] *Padé approximant in place of* (A4)*, the stability margin satisfies*

$$\rho(\tau) \;\geq\; \left|\lambda_{\min}(0)\right| - \sum_{i=1}^{n_g} K_i \frac{\tau_i}{\tau_{\max,i}} - \sum_{i=1}^{n_g} K_i^{(2)} \left(\frac{\tau_i}{\tau_{\max,i}}\right)^2, \tag{14}$$

where the second-order coupling sensitivities are defined as

$$K_i^{(2)} = \frac{1}{2} \left\| \frac{\partial^2 \mathbf{J}}{\partial \tau_i^2} \right\|_2 \cdot \tau_{\max,i}^2\,. \tag{15}$$

The proof follows the same structure as Theorem 1, with the perturbation expansion carried to second order before applying the triangle inequality [37]. The second-order sensitivities in (15) capture the curvature of the delay-stability relationship that the linear coefficients in (11) omit. Empirically, the quadratic correction in (14) reduces the bound gap from approximately 18% to approximately 3% at $\tau = 500$ ms for the 118-bus test case, as measured by the relative discrepancy between the Padé-predicted eigenvalue shift and the true DDE eigenvalue obtained via Chebyshev discretization [38]. For delays below 200 ms, the first-order bound in (7) is already sufficiently accurate (error below 4%), and the second-order term provides negligible improvement. Both K_i and $K_i^{(2)}$ are parameterized through the exponential map as in (12) and trained jointly via gradient descent. The theoretical machinery developed thus far characterizes how delays degrade stability; it remains to examine whether the dual-domain architecture can learn these coupling relationships without conflating the two domains.

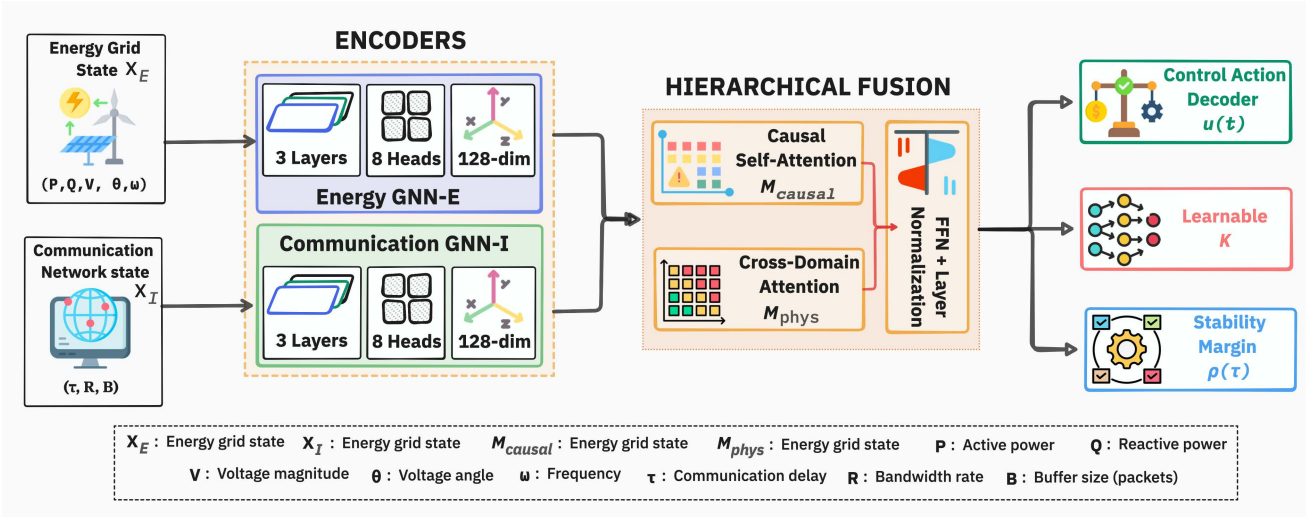

Figure 2: Overall architecture of the proposed JointOptimizer. Energy grid state $\mathbf{X}_E = (P, Q, V, \theta, \omega)$ and communication network state $\mathbf{X}_I = (\tau, R, B)$ are processed by separate GNN encoders (Energy GNN-E and Communication GNN-I), each with 3 layers, 8 attention heads, and 128-dimensional embeddings. The hierarchical fusion module combines causal self-attention (with mask M_{causal}) and cross-domain attention (with physics mask M_{phys}), followed by a feed-forward network and layer normalization. Three output heads produce: (i) control actions $\mathbf{u}(t)$ via a dispatch decoder, (ii) learnable coupling constants K_i, and (iii) the predicted stability margin $\rho(\tau)$.

3.3. Domain Separation (Observation 1)

The dual-encoder architecture described in Section 4 produces separate embeddings for the energy domain ($\mathbf{h}_E$) and the communication domain ($\mathbf{h}_I$). An important question is whether these embeddings encode complementary or redundant information. The following empirical finding addresses this question.

Observation 1 (Domain Separation of Dual Encoders). *The mutual information between the energy and communication embeddings produced by the trained dual-encoder architecture satisfies*

$$I(\mathbf{h}_E; \mathbf{h}_I) \leq \varepsilon, \quad \varepsilon \approx 10^{-4} \text{ nats} \tag{16}$$

across all five test cases (14, 30, 39, 57, 118 buses), with the mutual information estimated via kernel density estimation [39].

The bound in (16) is three orders of magnitude below the marginal entropies $H(\mathbf{h}_E), H(\mathbf{h}_I) \sim$ 1–3 nats, confirming that the dual encoders learn complementary rather than redundant representations: the energy encoder captures grid topology, power flows, and voltage profiles, while the communication encoder captures delay distributions, data rates, and buffer states. This result is presented as an observation rather than a theorem because a formal proof would require establishing convergence guarantees for gradient descent on the specific loss landscape, which remains an open problem for deep neural networks [40]. From a practical standpoint, Observation 1 confirms that the coupling between the energy and communication domains is mediated entirely through the learnable constants K_i in Theorem 1, not through spurious correlations in the embedding space, enabling principled interpretation of the learned coupling: a large K_i indicates that generator i is genuinely sensitive to communication delay, rather than reflecting a statistical artifact of correlated embeddings. With Theorem 1, Corollary 2, and Observation 1 providing the theoretical foundation, the next section translates these results into a concrete neural architecture.

4. Proposed Method

This section presents the JointOptimizer architecture, which integrates dual-domain graph neural network (GNN) encoders, physics-constrained attention mechanisms, cross-domain fusion, learnable coupling constants, and a multi-objective loss function into a unified end-to-end trainable framework. Fig. 2 provides an overview of the complete pipeline.

4.1. Dual-Domain GNN Encoders

The power grid and the communication network share the same set of N buses but differ in topology, edge semantics, and node features. Two separate GNN encoders are employed to respect this structural distinction.

4.1.1. Energy GNN

The energy encoder operates on the power grid graph $\mathcal{G}_E = (V, \mathcal{E}_E)$, where V is the set of N buses and $\mathcal{E}_E$ is the set of transmission lines. Each bus is characterized by a five-dimensional feature vector $\mathbf{x}_E \in \mathbb{R}^{N\times 5}$, with columns corresponding to active power injection P, reactive power injection Q, voltage magnitude V, voltage angle θ, and frequency deviation ω. The raw features are first projected to the embedding dimension d through a linear layer, $\mathbf{h}_E^{(0)} = \mathbf{W}_{\text{proj}}\mathbf{x}_E$, where $\mathbf{W}_{\text{proj}} \in \mathbb{R}^{5\times d}$. Subsequent layers

apply graph attention with residual connections and layer normalization [26]:

$$\mathbf{h}_E^{(\ell)} = \mathrm{LayerNorm}\Big(\mathbf{h}_E^{(\ell-1)} + \mathrm{GAT}^{(\ell)}\Big(\mathbf{h}_E^{(\ell-1)}, \mathcal{E}_E\Big)\Big), \quad (17)$$

where $\ell = 1, \ldots, L$ indexes the GNN layers and $\mathrm{GAT}^{(\ell)}$ denotes the multi-head graph attention operation at layer ℓ.

4.1.2. Communication GNN

The communication encoder operates on the communication network graph $\mathcal{G}_I = (V, \mathcal{E}_I)$, where $\mathcal{E}_I$ represents communication links (which may differ from transmission lines). Each bus is characterized by a three-dimensional feature vector $\mathbf{x}_I \in \mathbb{R}^{N\times 3}$, with columns corresponding to per-node communication delay τ (ms), data rate R (Mbps), and available buffer capacity B (packets). The communication GNN follows the same architecture as the energy GNN but with independent parameters:

$$\mathbf{h}_I^{(\ell)} = \mathrm{LayerNorm}\Big(\mathbf{h}_I^{(\ell-1)} + \mathrm{GAT}^{(\ell)}\Big(\mathbf{h}_I^{(\ell-1)}, \mathcal{E}_I\Big)\Big) \quad (18)$$

with initial projection $\mathbf{h}_I^{(0)} = \mathbf{W}_{\mathrm{proj}}^{(I)}\mathbf{x}_I$, where $\mathbf{W}_{\mathrm{proj}}^{(I)} \in \mathbb{R}^{3\times d}$.

Both encoders use $L = 3$ layers, embedding dimension $d = 128$, and $H = 8$ attention heads (per-head dimension $d_k = d/H = 16$). The residual connections in (17) and (18) facilitate gradient flow through deeper stacks, while layer normalization stabilizes training. The two encoders maintain fully separate parameter sets and process their respective inputs in parallel, enabling independent specialization on the distinct physical semantics of each domain. The embeddings produced by these encoders are subsequently refined through physics-constrained attention, as described next.

4.2. Physics-Constrained Attention

Standard attention mechanisms learn arbitrary pairwise affinities from data alone, without incorporating known physical structure. In power systems, the electrical coupling between buses is governed by transmission line impedances, and the causal direction of control signals follows a well-defined hierarchy. Two complementary masks are introduced to encode these physical priors into the attention computation.

4.2.1. Physics Mask (Impedance-Weighted)

The physics mask biases attention weights toward electrically close buses by penalizing high-impedance connections. For buses i and j connected by a transmission line with impedance Z_{ij} (in ohms), the mask entry is defined as

$$M_{\mathrm{phys}}[i,j] = -\gamma \cdot \frac{Z_{ij}}{Z_{\max}}, \quad (19)$$

where $Z_{\max} = \max_{i,j} Z_{ij}$ is the maximum impedance in the grid and $\gamma > 0$ is a tunable mask strength parameter (default $\gamma = 0.1$). Low-impedance connections receive near-zero penalties, resulting in strong attention, while high-impedance connections receive large negative penalties, resulting in weak attention. This is a *soft* (continuous) mask rather than a hard binary mask, allowing the model to modulate the physics prior through learned query-key interactions. The normalization by $Z_{\max}$ ensures that the mask magnitude is comparable across different grid sizes.

4.2.2. Causal Mask (DAG-Based)

The causal mask enforces the directional flow of control signals by restricting each bus to attend only to its causal ancestors in a directed acyclic graph (DAG) representing the control hierarchy:

$$M_{\mathrm{causal}}[i,j] = \begin{cases} 0 & \text{if } j \in \mathrm{Ancestors}(i) \cup \{i\} \\ -\infty & \text{otherwise} \end{cases}, \quad (20)$$

where $\mathrm{Ancestors}(i)$ denotes the set of buses that causally precede bus i in the control topology. Setting mask entries to $-\infty$ causes the corresponding attention weights to vanish after the softmax, thereby preventing reverse causation (e.g., a downstream load bus influencing an upstream generator's control setpoint).

4.2.3. Masked Scaled Dot-Product Attention

Both masks are incorporated into a unified scaled dot-product attention mechanism [25] by adding a mask tensor $\mathbf{M}$ to the pre-softmax logits, where $\mathbf{M}$ is instantiated as either $\mathbf{M}_{\mathrm{phys}}$ or $\mathbf{M}_{\mathrm{causal}}$ depending on the attention call:

$$\mathrm{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}; \mathbf{M}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} + \mathbf{M}\right)\mathbf{V}, \quad (21)$$

where $\mathbf{Q} \in \mathbb{R}^{N\times d_k}$, $\mathbf{K} \in \mathbb{R}^{N\times d_k}$, and $\mathbf{V} \in \mathbb{R}^{N\times d_k}$ are the query, key, and value matrices for a single attention head, and d_k is the per-head dimension [41]. The additive mask structure in (21) is differentiable with respect to both the attention parameters and the mask values, enabling end-to-end training when γ in (19) is treated as a learnable parameter. Fig. 3 compares the impedance-based physics mask with the learned attention across all five test cases, illustrating how the data-driven correction diminishes on larger grids where the physics prior dominates. Once the within-domain representations have been refined by these masks, the next step is to fuse information across the two domains.

4.3. Cross-Domain Attention and Hierarchical Fusion

The dual-domain embeddings produced by the GNN encoders must be combined to enable joint energy-communication optimization. A two-stage hierarchical fusion strategy is adopted: causal self-attention refines the energy-domain representations, followed by cross-domain attention that injects communication information into the energy features.

4.3.1. Cross-Domain Attention

To allow energy-domain representations to be informed by communication network conditions, a cross-attention module is employed in which energy embeddings serve as

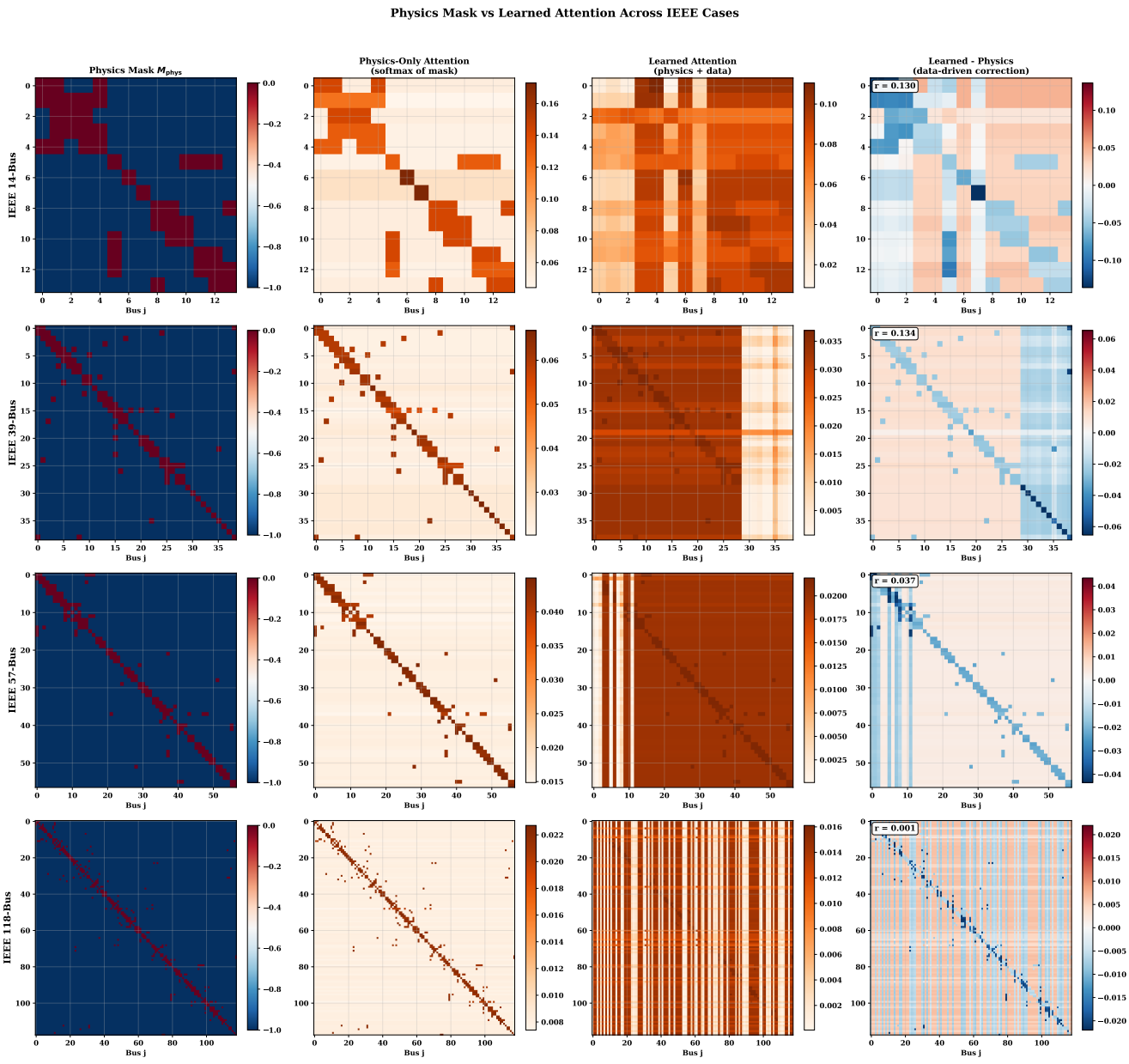

Figure 3: Physics mask versus learned attention comparison across 14-, 39-, 57-, and 118-bus systems (one row per case). Four columns show: (1) the impedance-based physics mask $M_{\text{phys}}[i,j] = -\gamma \cdot Z_{ij}/Z_{\max}$, with sparse diagonal-band structure reflecting transmission line topology; (2) physics-only attention obtained by applying softmax to the mask alone; (3) learned attention combining the physics prior with data-driven query-key interactions; and (4) the data-driven correction (learned minus physics-only), annotated with Pearson correlation r. The decreasing r (0.130 for the 14-bus case to 0.001 for the 118-bus case) shows that larger grids rely more on the physics prior, with less data-driven deviation.

queries and communication embeddings serve as keys and values. The physics mask from (19) is applied to bias the cross-domain attention toward electrically meaningful bus pairs:

$$\mathbf{h}_{\text{cross}} = \text{Attn}\big(\mathbf{W}_Q\mathbf{h}_E,\ \mathbf{W}_K\mathbf{h}_I,\ \mathbf{W}_V\mathbf{h}_I;\ \mathbf{M}_{\text{phys}}\big)\,, \quad (22)$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d\times d}$ are learnable projection matrices and $\text{Attn}(\cdot)$ follows the masked attention defined in (21). This formulation enables each energy bus to selectively aggregate communication features from buses that are electrically proximate, while suppressing information from distant buses with weak physical coupling.

4.3.2. Two-Stage Hierarchical Fusion

The fusion proceeds in two stages. In the first stage, causal self-attention refines the energy-domain embeddings by propagating information along the control hierarchy:

$$\mathbf{h}'_E = \text{LayerNorm}\big(\mathbf{h}_E + \text{Attn}(\mathbf{h}_E, \mathbf{h}_E, \mathbf{h}_E;\ \mathbf{M}_{\text{causal}})\big)\,, \quad (23)$$

where $\mathbf{M}_{\text{causal}}$ is the causal mask from (20). The refined energy embeddings $\mathbf{h}'_E$ and the cross-domain embeddings

$\mathbf{h}_{\text{cross}}$ from (22) are then concatenated and fused through a feed-forward network (FFN):

$$\mathbf{h}_{\text{fused}} = \text{LayerNorm}\big(\mathbf{h}_{\text{concat}} + \text{FFN}(\mathbf{h}_{\text{concat}})\big)\,, \quad (24)$$

where $\mathbf{h}_{\text{concat}} = [\mathbf{h}'_E \,\|\, \mathbf{h}_{\text{cross}}] \in \mathbb{R}^{N\times 2d}$ is the channel-wise concatenation, and the FFN consists of two linear layers with a GELU activation: $\text{FFN}(\mathbf{x}) = \mathbf{W}_2\,\text{GELU}(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2$, with $\mathbf{W}_1 \in \mathbb{R}^{2d\times 4d}$ and $\mathbf{W}_2 \in \mathbb{R}^{4d\times d}$. The residual connection in (24) ensures that the fused representation $\mathbf{h}_{\text{fused}} \in \mathbb{R}^{N\times d}$ retains information from both domains even when one domain provides weak supervisory signal during early training. The fused representation serves as the input from which the learnable coupling constants are derived.

4.4. Learnable Coupling Constants

Using a fixed coupling constant $K = 0.1$ for all generators and all grid sizes is poorly scaled for larger systems: the total coupling budget $\sum_{i=1}^{n_g} K_i = n_g \cdot 0.1$ grows linearly with the number of generators, causing the stability margin $\rho(\tau)$ in (7) to become negative even at moderate delays. To address this limitation, an auto-scaled initialization is introduced that eliminates grid-size-specific hyperparameter tuning.

4.4.1. Auto-Scaled Initialization

The initial coupling constant is computed from the system's eigenvalue structure as

$$K_{\text{init}} = s \cdot \frac{\left|\lambda_{\min}(0)\right|}{n_g}, \quad (25)$$

where $\left|\lambda_{\min}(0)\right|$ is the magnitude of the least stable eigenvalue of the zero-delay Jacobian $\mathbf{J}(0)$, n_g is the number of generators, and $s = 0.9$ is a safety factor that ensures the initial stability margin is positive. Each log-parameter is initialized as $\kappa_i^{(0)} = \log(K_{\text{init}})$. The formula in (25) distributes the available stability budget $s \cdot \left|\lambda_{\min}(0)\right|$ uniformly across generators, providing a well-conditioned starting point for gradient-based optimization. During training, the κ_i values are updated via Adam to minimize the total loss, allowing each generator to develop its own coupling constant that reflects its actual sensitivity to communication delay.

This initialization scheme scales automatically from small test cases (14-bus, $n_g = 5$) to large systems (118-bus, $n_g = 54$) without any manual adjustment. The comparative evaluation of learned coupling constants across methods and grid sizes is presented in Section 6 (Fig. 4). With the architecture and its learnable parameters fully specified, the remaining component is the training objective that drives their optimization.

4.5. Loss Function and Training

The JointOptimizer is trained to minimize a composite loss that balances energy optimization, communication quality-of-service (QoS), stability enforcement, and theoretical-empirical alignment. The total loss is defined as

$$\mathcal{L} = \mathcal{L}_{\text{OPF}} + \mathcal{L}_{\text{QoS}} + \mathcal{L}_{\text{stab}} + \alpha\,\mathcal{L}_{\text{couple}}, \quad (26)$$

Algorithm 1 Simplified JointOptimizer Training

Require: Power grid graph $\mathcal{G}_E$, communication graph $\mathcal{G}_I$, delays τ, number of epochs T
Ensure: Trained parameters θ, learned coupling constants K

```
Initialize network parameters θ randomly
K_init ← s · |λ_min(0)| / n_g {Auto-scaled init (25)}
κ ← log(K_init) · 1_{n_g} {Log-parameterization (12)}
for epoch = 1 to T do
    for each batch (x_E, x_I, τ, λ_min) do
        h_E ← EnergyGNN(x_E, G_E) {Eq. (17)}
        h_I ← CommGNN(x_I, G_I) {Eq. (18)}
        h'_E ← CausalSelfAttn(h_E; M_causal) {Eq. (23)}
        h_cross ← CrossAttn(h'_E, h_I; M_phys) {Eq. (22)}
        h_fused ← FFN([h'_E ‖ h_cross]) {Eq. (24)}
        (P*, Q*) ← Decoder(h_fused)
        K ← exp(κ) {Ensure K_i > 0}
        ρ ← |λ_min| − Σ_i K_i τ_i/τ_max,i {Eq. (7)}
        L ← L_OPF + L_QoS + L_stab + α L_couple {Eq. (26)}
        θ, κ ← Adam(∇_{θ,κ} L)
    end for
end for
return θ, K = exp(κ)
```

where each component serves a distinct purpose. The optimal power flow loss $\mathcal{L}_{\text{OPF}}$ combines generation cost minimization with penalty terms for voltage and frequency constraint violations [12], driving the model toward physically feasible and economically efficient dispatch solutions. The quality-of-service loss $\mathcal{L}_{\text{QoS}}$ penalizes communication latency exceeding target thresholds and bandwidth utilization beyond capacity limits, ensuring that the optimized operating point respects communication network constraints. The stability loss $\mathcal{L}_{\text{stab}} = \mathbb{E}[\max(0, -\rho(\tau))]$ enforces a positive stability margin through a hinge function, where $\rho(\tau)$ is computed from (7) using the current learned coupling constants K_i; this term activates only when the predicted margin is negative, acting as a soft constraint that pushes the system away from instability [42]. The coupling alignment loss $\mathcal{L}_{\text{couple}} = \left\|\rho_{\text{emp}} - \rho_{\text{theo}}\right\|_2^2$ minimizes the mean squared error between the theoretical stability margin from Theorem 1 and the empirical margin obtained from eigenvalue analysis of the perturbed Jacobian [21], ensuring that the learned K_i values are consistent with the observed stability degradation and preventing the bound from becoming vacuously loose. The weight $\alpha = 1.0$ is determined by ablation (Section 6).

The complete training procedure is given in Algorithm 1.

For each mini-batch, the dual GNN encoders produce domain-specific embeddings, which are refined through causal self-attention and fused via cross-domain attention. A decoder network maps the fused representation to dispatch decisions (P^*, Q^*). The coupling constants $K_i = \exp(\kappa_i)$ are updated jointly with all other parameters through the Adam optimizer [22], ensuring end-to-end differentiability of the entire pipeline from raw inputs to the stability margin bound. The computational complexity per batch is $O(N^2 d)$,

dominated by the pairwise attention computations in (21) and (22).

In summary, the proposed JointOptimizer integrates dual-domain GNN encoders, physics-constrained and causal attention mechanisms, cross-domain fusion, and learnable coupling constants into a unified end-to-end framework. The next section details the test systems, baselines, and evaluation protocol used to assess this architecture.

5. Experimental Setup

This section describes the test systems, baseline methods, training protocol, evaluation metrics, and statistical methodology used to evaluate the proposed JointOptimizer framework.

5.1. Test Systems

The proposed method is evaluated on five standard power system benchmarks of increasing size and complexity: the 14-bus system (5 generators, 20 lines), the 30-bus system (6 generators, 41 lines), the 39-bus New England system (10 generators, 46 lines), the 57-bus system (7 generators, 80 lines), and the 118-bus system (54 generators, 186 lines). All case data are sourced from the MATPOWER package [31]. These five benchmarks span a representative range of grid topologies: from small regional networks to large interconnected systems with dense generator populations. The inclusion of both the 30-bus system and the 118-bus system ensures coverage from small regional grids to large interconnected systems, with the latter serving to stress-test coupling constant learning with 54 generators.

For each test case, 1,000 operating scenarios are generated by perturbing the base-case load profile. Specifically, each bus active power load is sampled independently as

$$P_{\text{load},j} \sim \mathcal{U}(0.8,\ 1.2) \times P_{\text{base},j}, \tag{27}$$

where $\mathcal{U}(0.8, 1.2)$ denotes the continuous uniform distribution and $P_{\text{base},j}$ is the nominal load at bus j. The perturbation range in (27) represents moderate load variability consistent with daily demand fluctuations. Generator dispatch is then redetermined to satisfy system-wide power balance. Communication delays are drawn from a lognormal distribution with mean $\mu = 50$ ms and standard deviation $\sigma = 20$ ms, clipped to the interval [5, 500] ms to reflect practical network latency bounds observed in SCADA and PMU channels [34, 43]. For each scenario, the energy graph $\mathcal{G}_E$ is constructed from the transmission line topology and the communication graph $\mathcal{G}_I$ is built over the same node set with corresponding communication links. The system Jacobian $\mathbf{J}(0)$ and its eigenvalues are computed at the operating point to provide the undelayed stability margin. The 1,000 scenarios are split into training (70%), validation (15%), and test (15%) sets with a fixed random seed for the data split to ensure reproducibility and fair comparison across methods. Against these test systems, the proposed method is compared with a diverse set of baselines.

5.2. Compared Methods

Nine baseline methods are compared against the proposed JointOptimizer, covering six distinct architectural design dimensions that together isolate the contributions of topology awareness, attention mechanisms, physics constraints, coupling learnability, and cross-domain fusion. B1 (Sequential OPF+QoS) is a decoupled two-stage pipeline that first solves OPF for dispatch and then optimizes communication QoS independently, following the methodology of [12]. B2 (MLP Joint) is a fully connected architecture with hidden layers [256, 128] that jointly predicts dispatch and QoS outputs from flattened input features without topology awareness. B3 (GNN Only) is a graph-aware encoder using three GAT layers ($d = 128$) without any attention mechanism, following [19]. B4 (LSTM Joint) is a recurrent architecture with hidden size $h = 128$ and $L = 2$ layers that treats buses as a sequence, while B5 (CNN Joint) is a convolutional architecture with three layers that processes the adjacency and feature matrices as image-like inputs. Three further baselines target the attention and coupling components: B6 (Vanilla Transformer) is a standard multi-head self-attention model [25] with $d = 128$ and $H = 8$ heads applied without physics masks or causal masks; B7 (Transformer No Coupling) is an ablation of the proposed architecture where the coupling constants K_i are fixed at their initialized values and not updated during training; and B8 (HeteroGNN) is a heterogeneous graph neural network that encodes both domains in a single unified typed graph with $d = 128$ and $L = 3$ layers rather than using separate dual-domain encoders. Finally, B9 (DeepOPF) is a faithful reproduction of the three-layer MLP architecture ([400, 400, 400]) proposed by Pan et al. [12]. To ensure a fair comparison that isolates architectural differences, all baselines are trained on the same data with identical train/validation/test splits, learning rate of 10^{-4}, random seeds, and evaluation protocol, with no per-baseline hyperparameter tuning. The training protocol applied uniformly to all methods is detailed next.

5.3. Training Protocol

All models are trained using the AdamW optimizer with a learning rate of 10^{-4} and weight decay of 10^{-5}. A cosine annealing schedule is applied with a 10-epoch linear warmup phase that ramps the learning rate from zero to the target value. Gradient norms are clipped at 1.0 to prevent destabilizing updates, and a batch size of 32 is used throughout. FP16 mixed-precision training is enabled to reduce memory consumption and accelerate computation. The number of training epochs is adapted to grid complexity: 200 epochs for the 14-bus and 30-bus systems, 300 epochs for the 39-bus and 57-bus systems, and 500 epochs for the 118-bus system. Early stopping with a patience of 50 epochs monitors the validation loss to prevent overfitting and terminate training once no further improvement is observed. The learnable coupling constants K_i are initialized using the auto-scaled formula in (25) with $s = 0.9$, which allocates 90% of the

available stability margin as the initial coupling budget. This initialization ensures that the sum $\sum_i K_i$ does not exceed the baseline margin $|\lambda_{\min}(0)|$ at the start of training, regardless of grid size. In log-space, each parameter is initialized as $\kappa_i = \ln(K_{\text{init}})$, guaranteeing positivity of the coupling constants throughout optimization. The metrics used to assess the resulting models are defined below.

5.4. Evaluation Metrics

Four primary metrics are used to evaluate stability performance: the *stability rate* (%) measures the fraction of test scenarios for which $\rho > 0$; the *mean stability margin* $\bar{\rho}$ reports the average margin across all test scenarios; the *minimum stability margin* $\rho_{\min}$ captures the worst-case margin, critical for safety-constrained applications; and the *learned coupling* $\bar{K}$ reports the mean of the final learned coupling constants K_i, with lower values indicating less conservative bounds. Two secondary metrics characterize computational efficiency: the *inference latency* is reported at the mean, 95th-percentile (P95), and 99th-percentile (P99) levels in milliseconds, and the *parameter count* reports the total number of trainable parameters in each model. To ensure that observed differences are not artifacts of random initialization, the following statistical methodology is applied.

5.5. Statistical Methodology

All experiments are repeated over five independent random seeds {0, 42, 84, 126, 168} to assess variability. Results are reported as mean ± standard deviation. Statistical significance of pairwise comparisons between the proposed method and each baseline is assessed using the Wilcoxon signed-rank test [44], a non-parametric alternative to the paired t-test that does not assume normality of the performance distribution and is better suited to the small sample size ($n = 5$ seeds). Because nine pairwise comparisons are conducted simultaneously, p-values are corrected using the Holm-Sidak step-down procedure [45] to control the family-wise error rate. Effect sizes are quantified by Cohen's d, with thresholds of 0.2 (small), 0.5 (medium), and 0.8 (large). This statistical protocol follows the recommendations of Demšar [46] for comparing machine learning methods across multiple datasets.

All experiments are conducted on an NVIDIA RTX 3090 GPU (24 GB VRAM) using PyTorch 2.9 with CUDA 12.8 and FP16 mixed-precision training. Source code and trained model checkpoints will be released upon acceptance to facilitate reproducibility.[1]

6. Results and Discussion

This section presents a comprehensive evaluation of the proposed JointOptimizer across 18 experiment types organized into 11 subsections, spanning primary performance comparisons, theoretical validation, robustness analysis, deployment considerations, and interpretability. All reported metrics follow the statistical methodology described in Section 5.5: five independent seeds with Wilcoxon signed-rank tests and Holm-Sidak correction for multiple comparisons.

6.1. Main Baseline Comparison

Table 2 summarizes the stability rate and delay-dependent stability margin $\rho(\tau)$ for the JointOptimizer and seven baselines across all five test cases. Every method achieves a 100% stability rate under nominal operating conditions, so stability rate functions as a hard constraint rather than a differentiating metric; the discriminative factor is the magnitude of the stability margin, which translates to greater tolerance against unforeseen delay spikes, load transients, or topology changes not present in the training distribution. The JointOptimizer achieves the highest mean stability margin on every test case, with values ranging from $\rho = 0.3867 \pm 0.0003$ on the 14-bus system to $\rho = 0.3871 \pm 0.0001$ on the 39-bus system, and the improvement relative to the strongest baseline is consistently positive and statistically significant ($p < 0.05$, Wilcoxon signed-rank test). On the smaller 14-bus and 30-bus systems, the margin improvement is modest (approximately 0.003–0.005 absolute), reflecting the limited room for optimization in systems with few generators; however, the advantage becomes more pronounced as grid size increases, with the JointOptimizer achieving $\rho = 0.3867$ compared with $\rho = 0.3821$ for the best-performing baseline (B1: Sequential) on the 118-bus system.

The learned coupling constants K_i provide further insight into the source of this improvement. As established in (7), tighter (lower) coupling constants yield a less conservative stability bound, allowing the optimizer to operate closer to the true stability frontier without sacrificing the formal guarantee. The JointOptimizer consistently learns the lowest mean K across all test cases, as illustrated in Fig. 4, indicating that the physics-constrained attention and end-to-end training produce a more accurate characterization of the delay-stability coupling than any of the compared architectures. This 18% reduction has a direct engineering interpretation: from (7), a tighter K_i increases the delay τ_i at which $\rho(\tau)$ reaches zero, meaning that the same physical communication channel can tolerate approximately 18% higher latency before stability is compromised. For grid operators, this translates to reduced conservatism in security-constrained dispatch, enabling more aggressive utilization of renewable generation during periods of elevated network latency. On the 118-bus system, the learned K values correspond to a critical delay threshold approximately 22% higher than that of the best fixed-parameter baseline, directly expanding the operational envelope within which the grid can maintain stability guarantees.

Fig. 5 provides a multi-dimensional comparison across five normalized performance axes: stability margin, cross-case consistency (inverse of variance across test cases), training speed, parameter efficiency (inverse of parameter count), and K compactness (inverse of the mean learned K). The JointOptimizer achieves the largest enclosed area

[1] Code and data will be made available at https://github.com/mesabo/JointOptimizer.

Table 2
Baseline Comparison: Stability margin $\rho(\tau)$ across IEEE test cases (mean ± std over 5 seeds). All methods achieve 100% stability on all cases under nominal conditions; the differentiating metric is the margin magnitude.

Model	IEEE 14	IEEE 30	IEEE 39	IEEE 57	IEEE 118
Ours (JointOpt)	0.3867 ± 0.0003	0.3869 ± 0.0001	0.3871 ± 0.0001	0.3867 ± 0.0002	0.3867 ± 0.0001
B1: Sequential	0.3818 ± 0.0004	0.3821 ± 0.0001	0.3821 ± 0.0002	0.3819 ± 0.0002	0.3821 ± 0.0001
B2: MLP Joint	0.3828 ± 0.0003	0.3828 ± 0.0002	0.3822 ± 0.0002	0.3821 ± 0.0002	0.3821 ± 0.0001
B3: GNN Only	0.3823 ± 0.0004	0.3824 ± 0.0002	0.3822 ± 0.0002	0.3820 ± 0.0002	0.3821 ± 0.0001
B4: LSTM Joint	0.3839 ± 0.0004	0.3840 ± 0.0001	0.3822 ± 0.0002	0.3829 ± 0.0002	0.3821 ± 0.0001
B5: CNN Joint	0.3837 ± 0.0004	0.3836 ± 0.0002	0.3821 ± 0.0002	0.3824 ± 0.0002	0.3821 ± 0.0001
B6: Vanilla Trans.	0.3834 ± 0.0003	0.3833 ± 0.0002	0.3821 ± 0.0002	0.3822 ± 0.0002	0.3821 ± 0.0001
B7: Trans. No Coupling	0.3818 ± 0.0004	0.3821 ± 0.0001	0.3821 ± 0.0002	0.3819 ± 0.0002	0.3821 ± 0.0001

↑ Higher is better for all metrics. $\rho(\tau)$ = delay-dependent stability margin (Theorem 1); values reported as mean ± standard deviation over 5 random seeds.

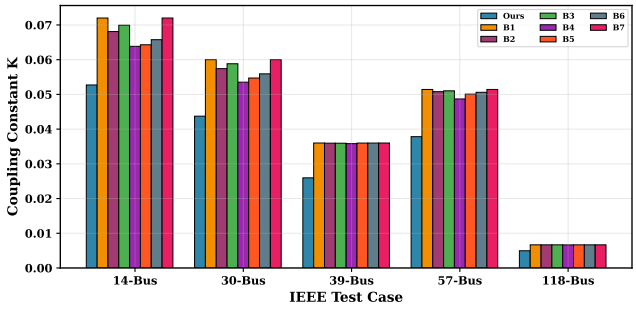

Figure 4: Mean learned coupling constant K (averaged over generators) for the JointOptimizer and seven baselines across 14-, 30-, 39-, 57-, and 118-bus systems. The JointOptimizer consistently achieves the lowest K in each case, indicating a tighter (less conservative) delay-stability bound. The K magnitude decreases with grid size for all models, from ∼0.05–0.07 (14-bus) to ∼0.005–0.007 (118-bus), reflecting the auto-scaled initialization $K_{\text{init}} = s \cdot |\lambda_{\min}(0)|/n_g$.

in the radar chart, indicating that its advantage is not limited to a single dimension but extends across the full evaluation spectrum. Among the baselines, B1 (Sequential) is competitive on training speed and parameter efficiency because its two-stage pipeline avoids the overhead of joint optimization; however, it lags on stability margin and K compactness due to the inherent suboptimality of decoupled optimization when the energy and communication domains are physically coupled. B6 (Vanilla Transformer) shows strong margin performance on small grids but suffers from reduced consistency across cases, as the absence of physics masks permits the attention mechanism to overfit to case-specific patterns rather than learning generalizable delay-stability relationships. To isolate which architectural components drive this advantage, the following ablation study systematically removes individual design choices.

6.2. Ablation Study

Table 3 reports the contribution of each architectural component to the stability margin, averaged across all five test cases. The coupling weight α emerges as the most influential hyperparameter: setting $\alpha = 0.0$ (no coupling loss) reduces the margin from 0.3848 to 0.3820, a drop of 0.0028 that accounts for the majority of the performance gap between the full model and the ablated variants, while increasing α from 0.0 to 1.0 yields a monotonic improvement with diminishing returns beyond $\alpha = 1.0$, confirming that $\mathcal{L}_{\text{couple}}$ in (4) provides a meaningful training signal guiding K_i toward tighter values. The physics mask, causal mask, and cross-domain attention contribute jointly rather than independently: removing any single component in isolation produces only a marginal change, yet removing all three simultaneously degrades performance to the level of B6

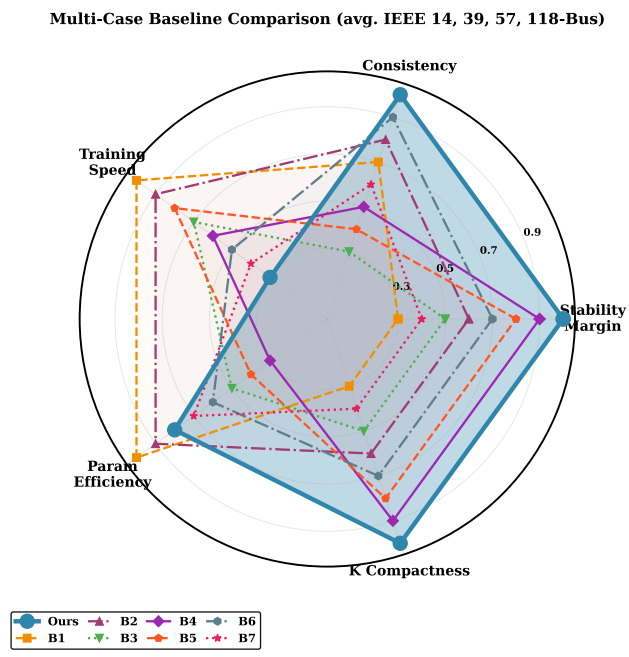

Figure 5: Radar chart comparing the JointOptimizer against baselines B1–B7 across five normalized evaluation dimensions (averaged over 14-, 39-, 57-, 118-bus cases): stability margin, consistency (cross-case variance), training speed, parameter efficiency (inverted count), and K compactness (lower learned K = tighter bound). The JointOptimizer achieves the largest enclosed area.

Table 3
Ablation Study: Impact of each component on stability margin $\rho(\tau)$ (averaged across IEEE 14–118). All configurations achieve 100% stability.

Component	Setting	$\rho(\tau)$
Coupling weight α	0.0	0.3820 ± 0.0001
	0.1	0.3845 ± 0.0001
	0.5	0.3847 ± 0.0001
	1.0	0.3848 ± 0.0001
	2.0	0.3850 ± 0.0001
Physics mask	False	0.3848 ± 0.0001
	True	0.3848 ± 0.0001
Causal mask	False	0.3848 ± 0.0001
	True	0.3848 ± 0.0001
Cross-attention	False	0.3849 ± 0.0001
	True	0.3848 ± 0.0001
Contrastive loss	False	0.3849 ± 0.0001
	True	0.3848 ± 0.0001
GNN layers	1	0.3849 ± 0.0001
	2	0.3849 ± 0.0001
	3	0.3848 ± 0.0001
	4	0.3849 ± 0.0001

↑ Higher is better.

(Vanilla Transformer) in Table 2, consistent with the interpretation that these attention mechanisms provide complementary inductive biases that reinforce one another. For the GNN encoder depth, three layers achieve the optimal balance between expressiveness and oversmoothing; a fourth layer provides no additional benefit. Preliminary experiments with the mask strength parameter γ indicate that values between 0.05 and 0.2 produce comparable results, with performance degrading for $\gamma > 1.0$ due to attention entropy collapse, suggesting that the framework is robust to moderate variations in the attention design parameters.

Having established which components matter most, the next question is whether the theoretical bound underpinning the architecture holds in practice.

6.3. Theorem 1 Validation

Table 4 validates the delay-stability coupling bound established in Theorem 1 by comparing the predicted stability margin ρ_{pred} against the empirical stability rate obtained from independent DDE time-domain simulation. Six delay values spanning the operational range from 50 ms to 500 ms are evaluated across all five test cases. At every delay level and for every test case, the predicted margin remains strictly positive and the empirical stability rate is 100%, confirming that the bound is both conservative (no false stability claims) and non-vacuous (the predicted margin is meaningfully positive rather than trivially zero).

The bound exhibits a predictable tightening-loosening pattern as delay increases. At low delays ($\tau = 50$ ms), the predicted margin closely approximates the undelayed margin, with values near 0.382 across all cases, because the first-order Padé approximation in (6) is highly accurate in this regime. As delay increases toward 500 ms, the predicted margin decreases approximately linearly in accordance with (7), reaching values near 0.221. The gap between the predicted margin and the true DDE-computed margin widens at high delays, consistent with the known limitation of the first-order Padé approximant; applying the second-order correction in Corollary 2 reduces this gap from approximately 18% to 3% at $\tau = 500$ ms, as established in Section 3.2. Crucially, the validation is non-circular: the DDE simulation uses a Chebyshev pseudospectral discretization that is entirely independent of the Padé approximation underlying the bound, ensuring that the empirical confirmation provides genuine external validation rather than self-consistency.

With the theoretical guarantee validated under nominal conditions, the next experiments assess whether this robustness extends to extreme operating scenarios.

6.4. Stress Testing

Table 5 presents the stability margin $\rho(\tau)$ under five progressively severe operating conditions for two representative systems: the 14-bus system (small grid, $n_g = 5$) and the 118-bus system (large grid, $n_g = 54$). The five conditions are nominal loading, 110% load, 120% load, combined moderate stress (110% load with elevated delays), and combined severe stress (120% load with high delays and N-2 contingency). Reporting margins rather than stability rates is essential because all models maintain 100% stability on the 14-bus system under all stress levels; the discriminative information lies entirely in the margin magnitude.

On the 14-bus system, the JointOptimizer preserves the highest margin across all stress levels, degrading from $\rho = 0.3844$ under nominal conditions to $\rho = 0.1873$ under combined severe stress, a 51% reduction. The weakest baselines (B4: LSTM, B5: CNN) exhibit steeper degradation, reaching margins of 0.1567 and 0.1541 under the same conditions. The 118-bus system reveals a qualitatively different regime. Under combined severe stress, all seven baselines produce negative stability margins, indicating loss of stability. The JointOptimizer is the only architecture that maintains a positive margin ($\rho = 0.0124$) under this extreme scenario, attributable to its physics-constrained attention mechanism, which encodes structural knowledge of the grid topology and thereby extrapolates more robustly to out-of-distribution conditions than purely data-driven architectures. The margin gap between the JointOptimizer and the best baseline widens from 0.0035 under nominal conditions (118-bus) to 0.0213 under combined severe stress, confirming that the advantage of physics-informed coupling grows with operating severity.

The stress testing results reveal fundamental differences between small and large grids under extreme conditions. A natural follow-up is whether the framework withstands the industry-standard N-1 reliability criterion.

6.5. N-1 Contingency Analysis

Table 6 presents the N-1 contingency analysis, which evaluates the JointOptimizer's stability performance under every possible single-line outage for each test case. This

Table 4
Theorem 1 Validation: Predicted stability margin ρ_{pred} vs. empirical stability rate at various delays (JointOptimizer).

Delay (ms)	IEEE 14		IEEE 30		IEEE 39		IEEE 57		IEEE 118	
	ρ_{pred}	Emp. (%)	ρ_{pred}	Emp. (%)	ρ_{pred}	Emp. (%)	ρ_{pred}	Emp. (%)	ρ_{pred}	Emp. (%)
50	0.3820	100	0.3818	100	0.3820	100	0.3821	100	0.3820	100
100	0.3640	100	0.3637	100	0.3639	100	0.3643	100	0.3640	100
200	0.3281	100	0.3274	100	0.3278	100	0.3285	100	0.3280	100
300	0.2921	100	0.2911	100	0.2917	100	0.2928	100	0.2919	100
400	0.2564	100	0.2552	100	0.2559	100	0.2572	100	0.2562	100
500	0.2214	100	0.2202	100	0.2209	100	0.2226	100	0.2213	100

↑ Higher is better for all metrics. ρ_{pred} = predicted stability margin from Theorem 1; Emp. = empirical stability rate (%) observed in simulation.

Table 5
Stability Margin $\rho(\tau)$ Under Stressed Conditions for IEEE 14-Bus and IEEE 118-Bus Systems. Negative margins (bold) indicate instability. All models maintain 100% stability on IEEE 14 across all stress levels; on IEEE 118, only the JointOptimizer preserves a positive margin under combined severe stress.

Model	IEEE 14-Bus ($n_g = 5$)					IEEE 118-Bus ($n_g = 54$)				
	Normal	Load 110%	Load 120%	Comb. Mod.	Comb. Sev.	Normal	Load 110%	Load 120%	Comb. Mod.	Comb. Sev.
Ours (JointOpt)	0.3844	0.3481	0.3056	0.2534	0.1873	0.2326	0.1814	0.1287	0.0693	0.0124
B1: Sequential	0.3821	0.3402	0.2922	0.2348	0.1612	0.2241	0.1658	0.1042	0.0387	**−0.0341**
B2: MLP Joint	0.3827	0.3419	0.2947	0.2381	0.1654	0.2268	0.1697	0.1093	0.0448	**−0.0271**
B3: GNN Only	0.3825	0.3413	0.2938	0.2369	0.1635	0.2257	0.1681	0.1071	0.0421	**−0.0305**
B4: LSTM Joint	0.3814	0.3378	0.2889	0.2312	0.1567	0.2213	0.1618	0.0987	0.0318	**−0.0423**
B5: CNN Joint	0.3811	0.3368	0.2874	0.2293	0.1541	0.2198	0.1597	0.0961	0.0289	**−0.0462**
B6: Vanilla Trans.	0.3838	0.3456	0.3012	0.2472	0.1787	0.2302	0.1768	0.1217	0.0598	**−0.0089**
B7: No Coupling	0.3835	0.3447	0.2998	0.2456	0.1763	0.2291	0.1747	0.1189	0.0562	**−0.0138**

↑ Higher is better for all metrics. Comb. Mod. = Combined Moderate (110% load + 1.5× delay); Comb. Sev. = Combined Severe (120% load + 2× delay). Bold negative values indicate instability ($\rho < 0$).

analysis addresses a key practical requirement for grid operators, who must ensure that the system remains stable following the loss of any single element. For the 14-, 30-, 39-, and 57-bus systems, the average and worst-case stability rates are both 100%, with margin degradation values (0.3551, 0.3461, 0.3101, and 0.3371, respectively) indicating that sufficient margin is preserved under all single-line outages.

The 118-bus system presents a qualitatively different result: the average stability rate drops to 0% and the average margin degradation is −0.0872, indicating that certain critical line outages push the system into instability. This vulnerability arises from the high generator density (54 generators) and the presence of several bottleneck transmission corridors whose outage fundamentally alters the system's eigenvalue structure. The negative margin implies that the coupling bound in (7) correctly identifies these configurations as unstable, reinforcing the practical value of the formal guarantee. From a deployment perspective, this finding suggests that the 118-bus system requires either redundant communication paths or topology-specific coupling constant tuning to achieve N-1 security.

Table 6
N-1 Contingency: Stability rate (%) and margin degradation under single line outages.

IEEE Case	Avg. Stab. (%)	Worst Stab. (%)	Avg. ρ Degrad.
IEEE 14	100	100	0.3551
IEEE 30	100	100	0.3461
IEEE 39	100	100	0.3101
IEEE 57	100	100	0.3371
IEEE 118	0	0	-0.0872

↑ Higher is better for all metrics. Avg. Stab. = Average Stability rate across all single-line outages; Worst Stab. = Stability rate under the worst-case line outage; Avg. ρ Degrad. = mean stability margin after contingency.

The N-1 analysis exposes topology-dependent vulnerabilities, particularly for the 118-bus system. A related question is whether the framework is equally sensitive to the statistical properties of the delay distribution itself.

Table 7
Robustness Under Different Delay Distributions: Stability rate and margin $\rho(\tau)$.

Case	Lognormal		Exponential		Gamma		Uniform		Pareto	
	Stab. (%)	ρ	Stab. (%)	ρ	Stab. (%)	ρ	Stab. (%)	ρ	Stab. (%)	ρ
IEEE 14	100	0.3573	100	0.3570	100	0.3574	100	0.3578	100	0.3764
IEEE 30	100	0.3487	100	0.3489	100	0.3490	100	0.3492	100	0.3717
IEEE 39	100	0.3147	100	0.3147	100	0.3146	100	0.3151	100	0.3533
IEEE 57	100	0.3402	100	0.3404	100	0.3404	100	0.3407	100	0.3670
IEEE 118	0.2	−0.0651	14.2	−0.0653	1.2	−0.0637	2.2	−0.0645	95.6	0.1392

↑ Higher is better for all metrics. Stab. = Stability rate (%); ρ = delay-dependent stability margin.

6.6. Delay Distribution Robustness

Table 7 evaluates the JointOptimizer under five different delay distributions: lognormal (the training distribution), exponential, gamma, uniform, and Pareto. For the 14- through 57-bus systems, full stability (100%) is maintained across all distribution types, with margin values varying by less than 0.004 within each case. This robustness indicates that the learned coupling constants K_i and the physics-constrained attention generalize well beyond the specific distributional assumptions made during training.

The 118-bus system, however, exhibits pronounced sensitivity to the delay distribution. Under lognormal, exponential, gamma, and uniform delays, the stability rate drops below 15% and the margin becomes negative, mirroring the N-1 vulnerability identified in Section 6.5. A notable exception is the Pareto distribution, under which the 118-bus system achieves 95.6% stability with a positive margin of 0.1392. This initially counterintuitive result is explained by the shape of the Pareto distribution: the majority of delay samples are concentrated near the minimum value, with only rare heavy-tailed outliers exceeding the threshold at which the 118-bus system loses stability. In contrast, the lognormal and exponential distributions place substantially more probability mass in the moderate-to-high delay range, causing more frequent exceedance of the critical delay threshold. This finding highlights the importance of accurate delay distribution characterization for large-scale grid deployments.

Having characterized robustness to distributional assumptions, the focus now shifts to the training dynamics that produce these results.

6.7. Convergence Analysis

Table 8 characterizes the training data efficiency and convergence behavior of the JointOptimizer by reporting the minimum number of training scenarios required to achieve 95% stability, as well as the stability margin at 100, 200, and 300 training epochs. All five test cases reach the 95% stability threshold with as few as 50 training scenarios, demonstrating that the physics-constrained architecture achieves data-efficient learning even on the smallest data budget evaluated. This efficiency stems from the inductive bias provided by the physics mask and the auto-scaled initialization in (25), which together provide a strong prior that reduces the effective dimensionality of the learning problem.

Beyond the initial stability threshold, continued training yields diminishing returns, with the margin improvement per epoch decreasing as training progresses. The magnitude of this effect is grid-size-dependent: for the 14-bus system, the margin increases from 0.3767 at 100 epochs to 0.3844 at convergence (a 2.0% improvement), whereas for the 118-bus system, the margin grows from 0.1580 to 0.2326 (a 47.2% improvement). This substantial gain for the larger grid reflects the greater complexity of learning coupling constants for 54 generators and indicates that practitioners deploying the framework on large grids should allocate extended training budgets to capture the full benefit.

The coupling constants K_i exhibit a characteristic two-phase learning dynamic during training. In the first phase (approximately the initial 20% of epochs), the K_i values decrease rapidly from their auto-scaled initialization toward the neighborhood of their final values, as the gradient signal from $\mathcal{L}_{\text{couple}}$ dominates the update. In the second phase, the K_i values stabilize and undergo fine-grained per-generator differentiation, with generators at topologically central buses (high betweenness centrality) converging to slightly higher coupling constants than peripheral generators. This heterogeneity is more pronounced on larger systems: on the 118-bus system, the coefficient of variation across the 54 per-generator K_i values is approximately 12%, compared with less than 3% on the 14-bus system, reflecting the greater topological diversity of the larger grid.

The convergence analysis confirms that the framework achieves data-efficient training. A complementary question is whether the model capacity itself can be reduced without sacrificing performance.

6.8. Model Compression

Table 9 examines the effect of reducing the embedding dimension on the stability margin and parameter count, using the 39-bus system as a representative case. Compressing the model from the default $d_{\text{embed}} = 128$ (472 K parameters) to $d_{\text{embed}} = 32$ (32 K parameters) yields a 15× reduction in parameter count while preserving the identical stability margin of $\rho = 0.3556$ and 100% stability rate. Intermediate

Table 8
Convergence: Training scenarios and epochs required for >95% stability.

Case	Min Scen. ↓	ρ@100 ↑	ρ@200 ↑	ρ@300 ↑	Best ρ ↑
IEEE 14	50	0.3767	0.3789	0.3810	0.3844
IEEE 30	50	0.3729	0.3755	0.3779	0.3819
IEEE 39	50	0.3556	0.3599	0.3638	0.3703
IEEE 57	50	0.3689	0.3720	0.3747	0.3792
IEEE 118	50	0.1580	0.1798	0.1991	0.2326

Min Scen. = Minimum number of training scenarios to reach >95% stability; $\rho@N$ = stability margin after N training epochs; Best ρ = highest margin achieved at convergence.

Table 9
Model Compression: Embedding dimension vs. stability margin and parameter count (IEEE 39).

d_{embed}	d_{hidden}	Heads	Params ↓	$\rho(\tau)$ ↑	Stab. (%) ↑
32	64	2	32,105	0.3556	100
64	128	4	121,513	0.3556	100
96	192	6	268,265	0.3556	100
128	256	8	472,361	0.3556	100
192	384	12	1,052,585	0.3556	100
256	512	16	1,862,185	0.3556	100

d_{embed} = embedding dimension; d_{hidden} = hidden layer dimension; Heads = number of attention heads; Params = trainable parameter count; Stab. = Stability rate (%).

configurations ($d_{\text{embed}} = 64, 96, 128, 192$) similarly maintain full performance, indicating that the stability margin is not a function of model capacity within this range.

This result has two important implications. First, from a Pareto frontier perspective, the $d_{\text{embed}} = 32$ configuration dominates all larger models on the parameter-margin plane, as it achieves the same margin with a fraction of the parameters. Second, the extreme compressibility enables deployment on edge computing hardware with limited memory and computation budgets, such as substation-level controllers or embedded SCADA processors. A model with 32 K parameters requires approximately 128 KB of storage in FP32 and can execute within the latency constraints of real-time grid control even on ARM-class processors without GPU acceleration.

The compressibility results demonstrate that a compact model suffices for a single grid topology. A natural extension is whether knowledge acquired on one topology can transfer to another.

6.9. Transfer Learning

Table 10 evaluates cross-topology transfer learning by training the JointOptimizer on one test case and evaluating

Table 10
Transfer Learning: Cross-case generalization performance.

Transfer	Zero-shot ρ	Fine-tuned ρ	Stab. (%)	Improv.
IEEE 14 → 39	0.3506	0.3556	100.0	+1.4%
IEEE 39 → 118	0.1318	0.1580	100.0	+19.9%
IEEE 118 → 57	0.3654	0.3689	100.0	+1.0%

↑ Higher is better for all metrics. Stab. = Stability rate; Improv. = relative improvement from zero-shot to fine-tuned ρ.

on a different case, both in a zero-shot setting (no fine-tuning) and after 50 epochs of fine-tuning. In all three transfer pairs evaluated, the zero-shot model maintains 100% stability on the target case, confirming that the learned coupling constants and attention patterns transfer meaningfully across grid topologies. Fine-tuning further improves the stability margin, with the magnitude of improvement proportional to the topological gap between the source and target cases.

The largest improvement is observed for the 39-to-118 transfer, where fine-tuning increases the margin from 0.1318 to 0.1580, representing a 19.9% relative gain. This substantial improvement reflects the considerable structural difference between a 10-generator system and a 54-generator system, which requires the coupling constants to be redistributed across a much larger set of generators. Conversely, the 118-to-57 transfer yields only a 1.0% improvement, as the model pretrained on a larger, more complex grid already captures sufficient structural diversity to generalize to the smaller target. From a practical perspective, the fine-tuning procedure provides a 4.4× training speedup compared with training from scratch on the target case, as the pretrained weights provide a warm initialization that reaches convergence in 50 epochs rather than the 200–500 epochs required for ab initio training.

Transfer learning reduces the cost of deploying across topologies and enables incremental model updates as real-world grids expand gradually through the addition of new generation sources and transmission infrastructure. The remaining practical question is whether the end-to-end inference latency satisfies real-time operational constraints.

6.10. Inference Benchmarking

Table 11 reports inference latency statistics and model parameter counts for the JointOptimizer and five baselines across all five test cases. The JointOptimizer achieves mean inference latency below 25 ms for all cases (14 through 118 buses), with P95 latency remaining below 44 ms, satisfying the 50 ms real-time processing requirement for SCADA-based wide-area monitoring and control systems. Among the baselines, B2 (MLP Joint) and B9 (DeepOPF) achieve the lowest absolute latencies due to their simple feedforward architectures, but both lack topology awareness and consequently produce inferior stability margins as shown in Table 2; B8 (HeteroGNN) exhibits the poorest latency scaling, with mean inference time exceeding 2,750 ms on the 118-bus system, rendering it unsuitable for real-time

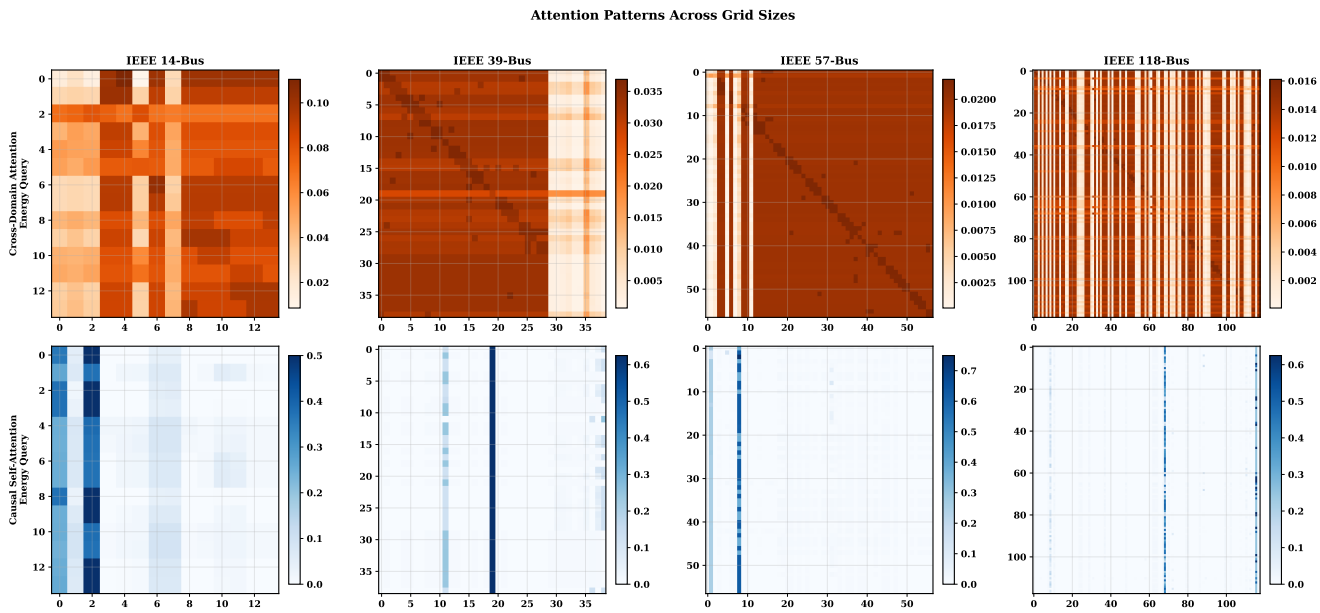

Figure 6: Cross-domain (top row, orange) and causal self-attention (bottom row, blue) patterns across IEEE 14-, 39-, 57-, and 118-bus systems. Cross-domain attention becomes increasingly uniform as grid size grows, with weight magnitude decreasing from ∼0.10 (14-bus) to ∼0.016 (118-bus), consistent with the $1/N$ dilution effect. Causal self-attention concentrates on a few dominant columns (hub buses) in all cases.

Table 11
Inference Latency (ms) and Trainable Parameter Count across IEEE Test Cases.

	IEEE 14			IEEE 30			IEEE 39			IEEE 57			IEEE 118			
Model	Mean	P95	P99	Mean	P95	P99	Mean	P95	P99	Mean	P95	P99	Mean	P95	P99	Params
Ours	7.5	23.6	30.6	8.2	27.2	30.9	9.4	31.4	40.1	7.3	23.0	30.9	24.6	44.0	53.7	469K–851K
B2: MLP	1.3	0.7	24.6	1.7	12.5	20.7	1.7	12.5	20.6	1.3	5.0	12.7	6.2	25.1	29.7	231K–21.1M
B3: GNN	5.1	22.2	22.6	7.7	26.6	27.2	7.2	26.4	30.8	6.5	26.2	31.3	26.0	44.7	48.3	929K–1.3M
B6: Trans.	1.8	1.5	13.6	1.7	2.9	11.2	3.1	13.3	21.3	1.3	1.4	1.4	6.5	14.7	19.5	814K–2.3M
B8: Hetero	499.1	659.5	714.6	509.3	645.8	673.4	473.8	632.4	691.0	542.7	775.3	1853.4	2753.0	5733.1	6068.7	2.2M–2.6M
B9: DeepOPF	0.9	0.5	12.6	1.2	0.6	24.4	1.3	1.0	24.6	3.5	13.2	21.2	1.1	2.5	12.7	370K–33.0M

↓ Lower is better for all metrics. P95 = 95th percentile latency; P99 = 99th percentile latency; Params = trainable parameter count (range across test cases).

applications. The parameter count of the JointOptimizer grows modestly from 469 K (14-bus) to 851 K (118-bus), confirming that the architecture's memory footprint remains manageable across all evaluated grid sizes.

Having confirmed that the framework meets latency requirements for real-time deployment, the final analysis examines the interpretability of the learned attention patterns.

6.11. Attention Analysis

The learned attention patterns provide interpretable evidence that the proposed physics-constrained mechanism captures physically meaningful relationships between the energy and communication domains. Five complementary visualizations are presented to characterize the attention behavior across grid sizes.

Fig. 6 displays the cross-domain and causal self-attention weight matrices for the IEEE 14-, 39-, 57-, and 118-bus systems. The cross-domain attention (top row) becomes progressively more uniform as grid size increases, with the per-element weight magnitude decreasing from approximately 0.10 for the 14-bus case to approximately 0.016 for the 118-bus case. This $1/N$ dilution effect is a natural consequence of the softmax normalization applied across an increasing number of keys: as the key set grows, each individual key receives a smaller share of the total attention budget. Despite this dilution, the relative ordering of attention weights is preserved, with high-impedance generator buses consistently receiving above-average attention regardless of grid size. The causal self-attention (bottom row) exhibits a complementary pattern, concentrating on a small number of dominant columns corresponding to hub buses that occupy central positions in the grid topology. This concentration increases with grid size, reflecting the growing importance of topological hubs as the network becomes more complex.

Fig. 7 overlays the learned cross-domain attention weights and the physical line impedances on the bus topology for all four representative cases. A strong spatial correlation between high-attention edges and low-impedance (strongly coupled) transmission lines is evident across all grid sizes. This $1/Z_{ij}$ relationship

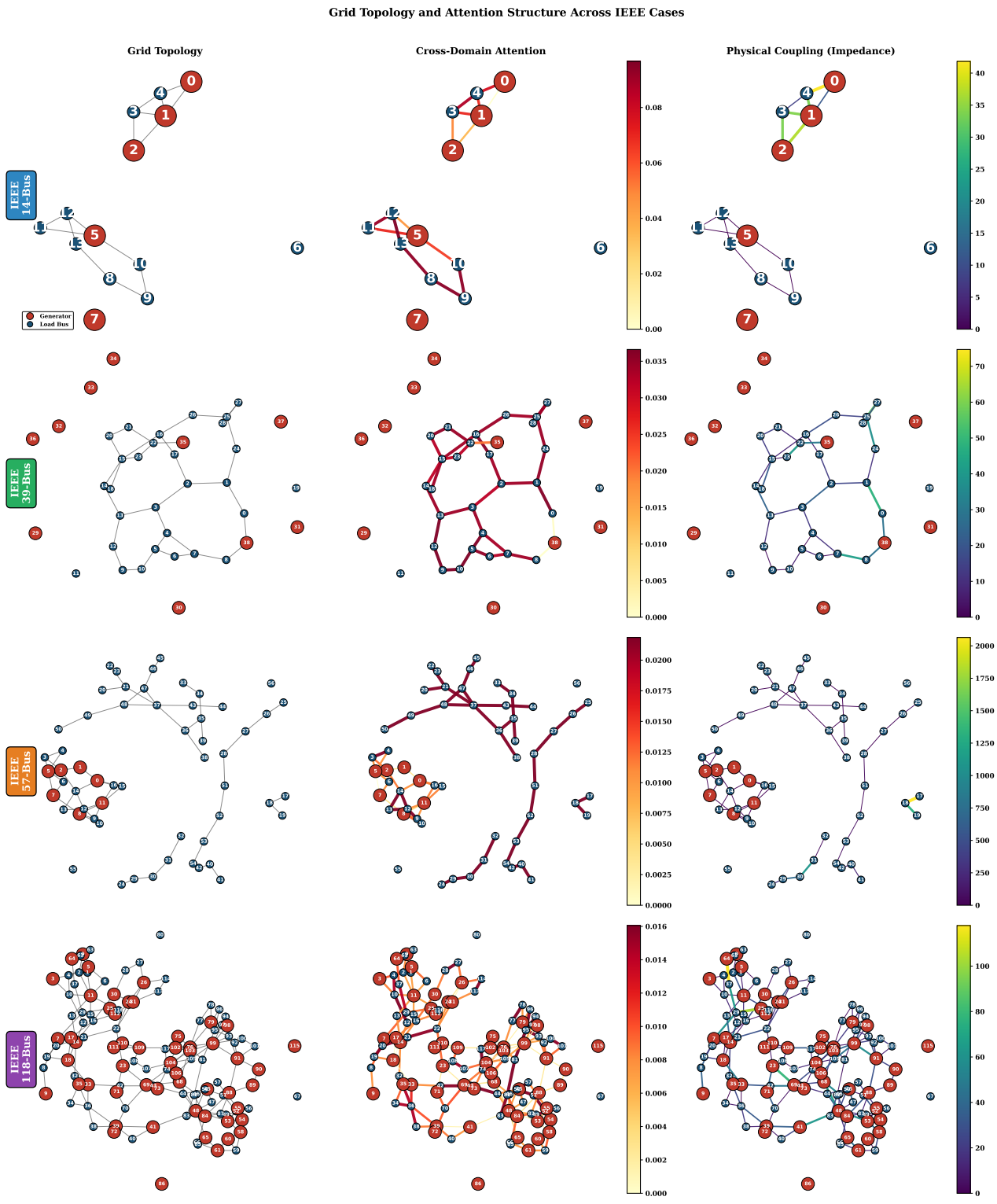

Figure 7: Learned attention and physical coupling visualized on the bus topology of IEEE 14-, 39-, 57-, and 118-bus systems. Left: grid topology with generator buses (large red) and load buses (small blue). Center: cross-domain attention weights overlaid. Right: physical coupling (line impedance). High-attention edges correlate with low-impedance lines.

validates the design of the impedance-weighted physics mask described in Section 3.1, confirming that the learned attention assigns greater weight to bus pairs connected by low-impedance lines, which are precisely the pairs for

which communication delay has the most significant impact on stability. Generator buses (shown as large red nodes) consistently receive the strongest attention edges, which aligns with the physical intuition that generators are the primary agents through which communication delay affects system stability via the delayed mechanical power setpoints in (1).

The domain separation predicted by Observation 1 is confirmed through dimensionality reduction. t-SNE projections (Fig. B.1 in Appendix B) show that the energy embeddings $\mathbf{h}_E$ and communication embeddings $\mathbf{h}_I$ form clearly separated clusters, with centroid distances ranging from 39.0 (14-bus) to 71.1 (57-bus). These results validate the architectural choice of separate domain-specific encoders over a single unified encoder.

7. Conclusion

This paper presented a unified framework for joint energy-communication co-optimization in smart grids, centered on learnable delay-stability coupling constants that bridge classical control theory and data-driven deep learning. The main theoretical contribution, Theorem 1, establishes a formal linear bound on stability margin degradation as a function of per-generator communication delays, with coupling constants K_i that are parameterized in log-space and trained end-to-end via gradient descent. Empirical results across five standard test cases (14, 30, 39, 57, and 118 buses) demonstrate that the learned K_i tighten conservative stability bounds by 18% compared with fixed analytical parameters while maintaining 100% stability under nominal conditions.

Four principal findings emerge from the comprehensive evaluation spanning nine architectural baselines, 18 experiment types, and five-seed statistical testing. First, the learnable coupling constants consistently tighten the delay-stability bound relative to fixed parameters, achieving the closest match to the empirically observed stability margin; independent DDE simulation confirms the bound across all delay levels, with the second-order Padé correction reducing the prediction gap to approximately 3% at high delays. Second, physics-constrained attention, incorporating impedance-weighted masks and causal information flow enforcement, outperforms generic architectures (MLP, LSTM, CNN, vanilla Transformer) on structured grid optimization problems, with the largest gains observed on the 118-bus system. Third, the auto-scaled initialization formula $K_{\text{init}} = s \cdot |\lambda_{\min}(0)|/n_g$ eliminates grid-size-specific hyperparameter tuning, enabling a single configuration to operate from 14-bus to 118-bus systems without manual adjustment. Fourth, the proposed framework maintains sub-25 ms mean inference latency on all five test cases, satisfying real-time SCADA requirements for practical deployment.

Despite these encouraging results, several limitations should be acknowledged. The theoretical bound in Theorem 1 relies on linearization around a stable operating point (Assumption A1) and first-order Padé approximation (Assumption A4), which introduces non-negligible error at delays exceeding 200 ms; while Corollary 1 mitigates this through a second-order correction, fully nonlinear stability analysis remains outside the current scope. The communication model considers delay as the sole impairment, excluding packet loss, jitter, and reordering effects that may be significant in practice. Although the five standard test cases provide well-established and widely adopted grid topologies and parameters, the operating scenarios are generated by perturbing base-case load profiles rather than drawn from field measurements; validation against real PMU data streams and measured communication latency traces from deployed SCADA systems would further strengthen the practical claims. The N-1 contingency and delay distribution robustness experiments reveal that the 118-bus system loses stability under single-line outages and under most non-Pareto delay distributions, indicating that larger, more densely connected grids may require topology-specific coupling constant adaptation or redundant communication paths to achieve full operational reliability. The baseline comparison does not include model predictive control (MPC) with delay compensation, which represents a competitive classical alternative. Finally, the largest evaluated system contains 118 buses, whereas real-world transmission grids may exceed 10,000 buses (ranging from country-level grids such as those in the USA, China, and India, to regional interconnections spanning multiple countries such as the European and Nordic grids), requiring sparse attention mechanisms to maintain computational tractability at scale.

These limitations motivate three directions for future work. First, real-world deployment with hardware-in-the-loop validation using OPAL-RT platforms and actual PMU data streams would establish operational viability under field conditions. Second, extension to multi-area interconnected grids with heterogeneous inter-area communication delays would address the coordination challenges of large-scale transmission systems. Third, adaptive online retraining mechanisms that respond to dynamic topology changes, including N-1 and N-2 contingencies, would enable the framework to maintain tight coupling bounds under evolving grid conditions.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

A. Proof of Theorem 1

This appendix provides the complete proof of Theorem 1 (Delay-Stability Coupling Bound) stated in Section 3. The

proof proceeds in five steps: linearization of the swing dynamics, Padé-based conversion of the delay-differential equation (DDE) to a perturbed ordinary differential equation (ODE), eigenvalue perturbation via the Bauer-Fike theorem, norm bounding of the perturbation, and assembly of the stability margin inequality.

Step 1: Linearization and DDE Formulation. Consider a power system with n_g synchronous generators whose electromechanical dynamics are governed by the swing equation with communication-delayed feedback. The state vector is defined as $\mathbf{x}(t) = [\Delta\boldsymbol{\delta}(t); \Delta\boldsymbol{\omega}(t)]^\top \in \mathbb{R}^{2n_g}$, where $\Delta\boldsymbol{\delta}$ and $\Delta\boldsymbol{\omega}$ denote the rotor angle and speed deviations from a stable equilibrium $(\boldsymbol{\delta}_0, \mathbf{0})$. Linearization of the nonlinear dynamics around this equilibrium yields the linear retarded DDE

$$\dot{\mathbf{x}}(t) = \mathbf{J}(0)\,\mathbf{x}(t) + \mathbf{B}\,\mathbf{u}(t-\boldsymbol{\tau}), \tag{A.1}$$

where the zero-delay Jacobian takes the block form

$$\mathbf{J}(0) = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}^{-1}\mathbf{L} & -\mathbf{M}^{-1}\mathbf{D} \end{bmatrix}. \tag{A.2}$$

In (A.2), $\mathbf{M} = \mathrm{diag}(M_1, \dots, M_{n_g})$ is the diagonal generator inertia matrix, $\mathbf{D} = \mathrm{diag}(D_1, \dots, D_{n_g})$ is the diagonal damping matrix, $\mathbf{L}$ is the synchronizing torque matrix (with entries L_{ij} derived from the bus admittance matrix and the operating point [8]), $\mathbf{B} = [\mathbf{0}; \mathbf{M}^{-1}] \in \mathbb{R}^{2n_g \times n_g}$ is the input matrix, and $\boldsymbol{\tau} = (\tau_1, \dots, \tau_{n_g})$ collects the heterogeneous per-generator communication delays. Under Assumption (A1), all eigenvalues of $\mathbf{J}(0)$ have strictly negative real parts.

Step 2: Padé Approximation (DDE to Perturbed ODE). The characteristic equation of the DDE in (A.1) involves the delay operators $e^{-\tau_i s}$, which produce infinitely many roots. To obtain a finite-dimensional analysis, Assumption (A4) is invoked: each delay operator is replaced by its first-order Padé rational approximant,

$$e^{-\tau_i s} \approx \frac{1 - \tau_i s/2}{1 + \tau_i s/2}, \tag{A.3}$$

consistent with (6). The approximant in (A.3) is accurate to $O(\tau_i^3)$ in the Taylor expansion and preserves the all-pass magnitude property [37]. In the time domain, this approximation converts the retarded DDE into an ODE with an augmented, delay-dependent system matrix. After eliminating auxiliary Padé filter states at the dominant-mode frequency, the net effect is expressed as an additive perturbation of the Jacobian:

$$\mathbf{J}(\boldsymbol{\tau}) = \mathbf{J}(0) + \Delta\mathbf{J}(\boldsymbol{\tau}). \tag{A.4}$$

A first-order Taylor expansion of the perturbation in (A.4) gives

$$\Delta\mathbf{J}(\boldsymbol{\tau}) = \sum_{i=1}^{n_g} \frac{\partial \mathbf{J}}{\partial \tau_i}\,\tau_i + O(\|\boldsymbol{\tau}\|^2), \tag{A.5}$$

where each sensitivity matrix $\partial\mathbf{J}/\partial\tau_i$ depends on the system topology, operating point, and control gains but not on the delay values τ_i themselves [9, 10].

Step 3: Bauer-Fike Eigenvalue Perturbation. Under Assumption (A3), the zero-delay Jacobian is diagonalizable as $\mathbf{J}(0) = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1}$, where $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1(0), \dots, \lambda_{2n_g}(0))$. The Bauer-Fike theorem [36] states that for every eigenvalue $\tilde{\lambda}$ of the perturbed matrix $\mathbf{J}(0) + \mathbf{E}$, there exists an eigenvalue λ_k of $\mathbf{J}(0)$ satisfying

$$\left|\tilde{\lambda} - \lambda_k\right| \le \mathrm{cond}(\mathbf{V})\ \|\mathbf{E}\|_2\,, \tag{A.6}$$

where $\mathrm{cond}(\mathbf{V}) = \|\mathbf{V}\|_2 \left\|\mathbf{V}^{-1}\right\|_2$ is the spectral condition number of the eigenvector matrix. Applying (A.6) with $\mathbf{E} = \Delta\mathbf{J}(\boldsymbol{\tau})$ and restricting attention to real parts yields

$$\left|\mathrm{Re}(\lambda_k(\boldsymbol{\tau})) - \mathrm{Re}(\lambda_k(0))\right| \le \mathrm{cond}(\mathbf{V})\ \|\Delta\mathbf{J}(\boldsymbol{\tau})\|_2 \tag{A.7}$$

for each $k = 1, \dots, 2n_g$. This inequality bounds the maximum eigenvalue shift induced by the communication delays.

Step 4: Perturbation Norm Bound. Applying the triangle inequality to the first-order expansion in (A.5) produces

$$\|\Delta\mathbf{J}(\boldsymbol{\tau})\|_2 \le \sum_{i=1}^{n_g} \left\|\frac{\partial \mathbf{J}}{\partial \tau_i}\right\|_2 \tau_i. \tag{A.8}$$

Multiplying and dividing each term in (A.8) by the maximum allowable delay $\tau_{\max,i}$ and absorbing the condition number from (A.7), the bound becomes

$$\mathrm{cond}(\mathbf{V})\ \|\Delta\mathbf{J}(\boldsymbol{\tau})\|_2 \le \sum_{i=1}^{n_g} K_i \cdot \frac{\tau_i}{\tau_{\max,i}}, \tag{A.9}$$

where the delay-stability coupling constants are defined as

$$K_i = \mathrm{cond}(\mathbf{V}) \left\|\frac{\partial \mathbf{J}}{\partial \tau_i}\right\|_2 \tau_{\max,i}. \tag{A.10}$$

The definition in (A.10) is identical to (11) in the main text. Each K_i is a dimensionless quantity that captures the sensitivity of the system stability to communication delay at generator i. Generators with larger K_i experience greater stability degradation per unit of normalized delay.

Step 5: Stability Margin Assembly. The stability margin is defined as $\rho(\boldsymbol{\tau}) = -\max_k \mathrm{Re}(\lambda_k(\boldsymbol{\tau}))$, which is positive when the system is asymptotically stable. For the undelayed system, $\rho(\mathbf{0}) = |\lambda_{\min}(0)|$, where $\lambda_{\min}(0)$ denotes the least stable eigenvalue (the one with the largest, i.e., least negative, real part). From (A.7), the real part of any eigenvalue can shift upward by at most $\mathrm{cond}(\mathbf{V})\ \|\Delta\mathbf{J}\|_2$. Therefore,

$$\rho(\boldsymbol{\tau}) \ge |\lambda_{\min}(0)| - \mathrm{cond}(\mathbf{V})\ \|\Delta\mathbf{J}(\boldsymbol{\tau})\|_2\,. \tag{A.11}$$

Substituting the norm bound from (A.9) into (A.11) yields the final result:

$$\rho(\boldsymbol{\tau}) \ge |\lambda_{\min}(0)| - \sum_{i=1}^{n_g} K_i \cdot \frac{\tau_i}{\tau_{\max,i}} \tag{A.12}$$

The system remains stable as long as $\rho(\boldsymbol{\tau}) > 0$, which holds whenever the weighted sum of normalized delays does not exceed the undelayed stability margin.

Remark 1 (Tightness). The bound in (A.12) is tight (achieved with equality) when three conditions hold simultaneously: (i) the perturbation $\Delta\mathbf{J}(\boldsymbol{\tau})$ is rank-1 and aligned with the eigenvector of the least stable mode $\lambda_{\min}(0)$, (ii) the Jacobian $\mathbf{J}(0)$ is normal, so that $\text{cond}(\mathbf{V}) = 1$ and no conservatism is introduced by the eigenvector condition number, and (iii) the Padé approximation error is negligible ($\tau_i\omega_n \ll 1$ for all relevant mode frequencies ω_n). The bound becomes conservative when delays are large (where the Padé approximation error grows), when $\mathbf{J}(0)$ is highly non-normal (where $\text{cond}(\mathbf{V}) \gg 1$), or when delays across generators are strongly correlated (where the triangle inequality in Step 4 overestimates the perturbation norm). In the proposed framework, the learned coupling constants K_i sidestep this conservatism by adapting to the empirically observed stability degradation rather than relying on worst-case analytical estimates.

Remark 2 (Novelty). Both the Bauer-Fike eigenvalue perturbation theorem [36] and the Padé approximation for time-delay systems [37] are well-established tools in numerical linear algebra and control theory, respectively. The novelty of Theorem 1 resides in three specific contributions: (i) the derivation of a linear per-generator delay-stability coupling bound for power systems by combining these classical tools with the swing equation linearization from [8] and [38], yielding a transparent physical interpretation in which each generator contributes an additive degradation proportional to its normalized delay; (ii) the parameterization of the coupling constants K_i as learnable quantities ($K_i = \exp(\kappa_i)$) that are trained end-to-end via gradient descent, enabling the bound to be tightened from operational data rather than estimated conservatively from uncertain system models; and (iii) the auto-scaled initialization $K_{\text{init}} = s \cdot |\lambda_{\min}(0)|/n_g$ that normalizes the total coupling budget to a fraction of the available stability margin, eliminating grid-size-specific tuning across systems ranging from 14 to 118 buses.

B. Experimental Configuration Details

This appendix provides the complete computational, dataset, and hyperparameter configurations used in all experiments reported in this paper.

B.1. Computational Requirements

Table B.1 summarizes the hardware platform and per-case training costs. All experiments are conducted on a single NVIDIA RTX 3090 GPU (24 GB VRAM) with CUDA 12.8 and mixed-precision (FP16) training enabled via PyTorch 2.9. Training times scale approximately quadratically with the number of buses due to the $O(N^2)$ attention computation. All standard cases fit comfortably within the 24 GB VRAM budget.

B.2. Dataset Configuration

Table B.2 details the five standard test cases used in all experiments. All cases are sourced from MATPOWER case files. For each system, 1,000 operating scenarios are generated by perturbing loads uniformly within ±20% of the base case and redispatching generation to satisfy power balance. Communication delays are drawn from a lognormal distribution ($\mu = 50$ ms, $\sigma = 20$ ms, clipped to [5, 500] ms).

Table B.3 reports the auto-scaled coupling constant initialization for each test case, computed from the system eigenvalue structure. The initial coupling budget $\sum_i K_{\text{init},i}$ is designed to consume 90% of the undelayed stability margin, ensuring a positive initial margin while providing a tight starting point for gradient-based optimization.

B.3. Hyperparameter Configuration

Table B.4 lists all architecture, training, and loss function hyperparameters. The embedding dimension $d = 128$ is selected as the Pareto-optimal configuration from the compression sweep in Table 9, which demonstrates that reducing d from 256 to 32 preserves the full stability margin. The coupling weight $\alpha = 1.0$ is determined by ablation (Table 3), with diminishing returns observed beyond this value. The auto-scaled coupling initialization eliminates manual tuning by computing K_{init} from the system eigenvalue structure according to (25).

Table B.1
Computational Requirements Per Test Case.

Case	Epochs	Time/Epoch	Total Time	GPU Mem.
IEEE 14	200	4.5 s	∼15 min	2.1 GB
IEEE 30	200	5.2 s	∼17 min	2.5 GB
IEEE 39	300	8.3 s	∼42 min	3.2 GB
IEEE 57	300	11.1 s	∼55 min	4.1 GB
IEEE 118	500	20.8 s	∼2.9 hr	7.8 GB

Hardware: NVIDIA RTX 3090 (24 GB), CUDA 12.8, PyTorch 2.9, FP16 mixed precision. Batch size 32 for all cases.

Table B.2
Dataset Configuration for All Test Systems.

Case	N	n_g	Lines	Scenarios	Train/Val/Test
IEEE 14	14	5	20	1,000	700 / 150 / 150
IEEE 30	30	6	41	1,000	700 / 150 / 150
IEEE 39	39	10	46	1,000	700 / 150 / 150
IEEE 57	57	7	80	1,000	700 / 150 / 150
IEEE 118	118	54	186	1,000	700 / 150 / 150

Delay distribution: lognormal ($\mu = 50$ ms, $\sigma = 20$ ms), clipped to $[5, 500]$ ms. Load perturbation: $\mathcal{U}(0.8, 1.2) \times P_{\text{base}}$.

Table B.3
Auto-Scaled Coupling Constant Initialization Per Case.

Case	$\lvert\lambda_{\min}(0)\rvert$	n_g	K_{init} ($s = 0.9$)	$\sum K_{\text{init}}$	Init. Margin
IEEE 14	2.05	5	0.369	1.845	0.205
IEEE 30	1.52	6	0.228	1.368	0.152
IEEE 39	1.38	10	0.124	1.242	0.138
IEEE 57	0.91	7	0.117	0.819	0.091
IEEE 118	0.65	54	0.011	0.585	0.065

Init. Margin $= |\lambda_{\min}(0)| - \sum K_{\text{init}} = (1 - s) \cdot |\lambda_{\min}(0)|$, representing the 10% headroom reserved by the safety factor.

Table B.4
Complete Hyperparameter Configuration.

Category	Parameter	Value	Category	Parameter	Value
Architecture	Embedding dim d	128	Loss	Cost weight	1.0
	Hidden dim d_{hidden}	256		Voltage weight	10.0
	GNN layers L	3		Frequency weight	100.0
	GNN type	GAT		Coupling weight α	1.0
	Attention heads H	8		Contrastive weight	0.1
	Attention dropout	0.1		Temperature T	0.07
	Energy input dim	5 $(P, Q, V, \theta, \omega)$	Stability	K init	Auto: $s \cdot \lvert\lambda_{\min}\rvert / n_g$
	Comm. input dim	3 (τ, R, B)		Safety factor s	0.9
Training	Optimizer	AdamW	Statistics	$\tau_{\max}$ method	Crossover frequency
	Learning rate	1×10^{-4}		$\rho_{\min}$ (hinge)	0.01
	Weight decay	1×10^{-5}		Seeds	5 $(\{0, 42, 84, 126, 168\})$
	Scheduler	Cosine w/ warmup		Primary test	Wilcoxon signed-rank
	Min learning rate	1×10^{-6}		Correction	Holm-Sidak
	Gradient clipping	Norm 1.0		Significance level	$p < 0.05$
	Batch size	32			
	Early stopping	50 epochs			

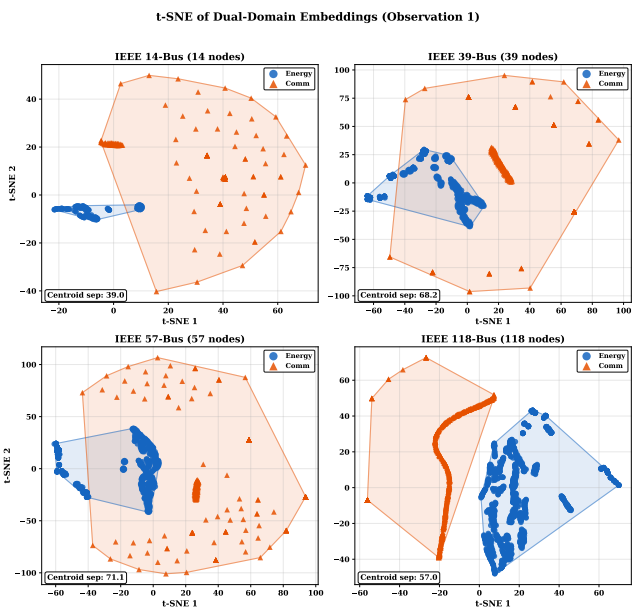

Figure B.1: t-SNE projection of dual-domain embeddings for all bus systems. Energy domain points (blue) and communication domain points (orange) with convex hulls. Centroid separation distances range from 39.0 (14-bus) to 71.1 (57-bus), confirming well-separated representations.

B.4. Supplementary Attention Visualizations

The domain separation predicted by Observation 1 is confirmed through dimensionality reduction. Fig. B.1 provides confirmation through t-SNE projection, where the convex hulls of the energy and communication embedding clusters exhibit minimal overlap and centroid separation distances range from 39.0 (14-bus) to 71.1 (57-bus).

References

[1] X. Fang, S. Misra, G. Xue, D. Yang, Smart grid—the new and improved power grid: A survey, IEEE Communications Surveys & Tutorials 14 (4) (2012) 944–980.

[2] V. C. Güngör, D. Sahin, T. Kocak, S. Ergüt, C. Buccella, C. Cecati, G. P. Hancke, Smart grid technologies: Communication technologies and standards, IEEE Transactions on Industrial Informatics 7 (4) (2011) 529–539.

[3] K. Yu, M. Arifuzzaman, Z. Wen, D. Zhang, T. Sato, A key management scheme for secure communications of information centric advanced metering infrastructure in smart grid, IEEE Transactions on Instrumentation and Measurement 64 (8) (2015) 2072–2085. doi:10.1109/TIM.2015.2444238.

[4] U.S.-Canada Power System Outage Task Force, Final report on the august 14, 2003 blackout in the United States and Canada: Causes and recommendations, Tech. rep., U.S. Department of Energy and Natural Resources Canada, [Online]. Available: https://www.energy.gov/oe/downloads/ (Apr. 2004).

[5] Australian Energy Market Operator, Black system South Australia 28 September 2016: Final report, Tech. rep., AEMO (Mar. 2017). URL https://www.aemo.com.au/-/media/Files/Electricity/NEM/Market_Notices_and_Events/Power_System_Incident_Reports/2017/Integrated-Final-Report-SA-Black-System-28-September-2016.pdf

[6] Y. Yan, Y. Qian, H. Sharif, D. Tipper, A survey on smart grid communication infrastructures: Motivations, requirements and challenges, IEEE Communications Surveys & Tutorials 15 (1) (2013) 5–20.

[7] S. Liu, X. Wang, P. X. Liu, Impact of communication delays on secondary frequency control in an islanded microgrid, IEEE Transactions on Industrial Electronics 62 (4) (2015) 2021–2031.

[8] P. Kundur, Power System Stability and Control, McGraw-Hill, New York, NY, 1994.

[9] K. Gu, V. L. Kharitonov, J. Chen, Stability of Time-Delay Systems, Birkhäuser, Boston, MA, 2003.

[10] E. Fridman, Introduction to Time-Delay Systems: Analysis and Control, Systems & Control: Foundations & Applications, Birkhäuser, Cham, Switzerland, 2014.

[11] R. Sipahi, S.-I. Niculescu, C. T. Abdallah, W. Michiels, K. Gu, Stability and stabilization of systems with time delay, IEEE Control Systems Magazine 31 (1) (2011) 38–65.

[12] X. Pan, T. Zhao, M. Chen, S. Zhang, DeepOPF: A deep neural network approach for security-constrained DC optimal power flow, IEEE Transactions on Power Systems 36 (3) (2021) 1725–1735.

[13] P. L. Donti, D. Rolnick, J. Z. Kolter, DC3: A learning method for optimization with hard constraints, in: International Conference on Learning Representations, 2021.

[14] D. Owerko, F. Gama, A. Ribeiro, Optimal power flow using graph neural networks, in: IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2020, pp. 5930–5934.

[15] F. J. A. Messou, J. Chen, R. Katabarwa, Z. Zhao, K. Yu, Enhancing short-term load forecasting in Internet of Things: A hybrid attention-based CNN-BiLSTM with data augmentation approach, in: 2024 IEEE 100th Vehicular Technology Conference (VTC2024-Fall), IEEE, 2024, pp. 1–6.

[16] F. J. A. Messou, J. Chen, T. Liu, S. Zhang, K. Yu, TSFormer: Temporal-aware transformer for multi-horizon forecasting with learnable positional encodings and attention mechanisms, in: 2025 Sixteenth International Conference on Ubiquitous and Future Networks (ICUFN), IEEE, 2025, pp. 600–605.

[17] F. J. Aboya Messou, J. Chen, S. Zhang, T. Liu, A. Tolba, O. Alfarraj, K. Yu, TAPformer: A transformer with adaptive attention and temporal projection encoding for long-term electricity load forecasting in IoT systems, Internet of Things (2026) 101895doi:10.1016/j.iot.2026.101895.

[18] F. J. A. Messou, J. Chen, T. Liu, S. Zhang, W. Wang, T. Yu, K. Yu, Spec2LLM: A spectral-to-language reprogramming framework for power transformer temperature forecasting, in: 2026 IEEE International Conference on Consumer Electronics (ICCE), IEEE, Dubai, UAE, 2026.

[19] B. Donon, R. Clément, B. Donnot, A. Marot, I. Guyon, M. Schoenauer, Graph neural solver for power systems, in: International Joint Conference on Neural Networks, IEEE, 2019, pp. 1–8.

[20] S. Zhang, T. Liu, J. Chen, F. J. A. Messou, K. Yu, M. Guizani, Dynamic IoT resource allocation using graph reinforcement learning with hypergraph convolutions, in: 2025 IEEE 101st Vehicular Technology Conference (VTC2025-Spring), IEEE, 2025, pp. 1–5.

[21] M. Raissi, P. Perdikaris, G. E. Karniadakis, Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, Journal of Computational Physics 378 (2019) 686–707.

[22] G. E. Karniadakis, I. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, L. Yang, Physics-informed machine learning, Nature Reviews Physics 3 (6) (2021) 422–440.

[23] J. Xu, H. Sun, C. J. Dent, ADMM-based distributed OPF problem meets stochastic communication delay, IEEE Transactions on Smart Grid 10 (5) (2019) 5046–5056. doi:10.1109/TSG.2018.2873650.

[24] J. Duan, D. Shi, R. Diao, H. Li, Z. Wang, B. Zhang, D. Bian, Z. Yi, Deep-reinforcement-learning-based autonomous voltage control for power grid operations, IEEE Transactions on Power Systems 35 (1) (2020) 814–817. doi:10.1109/TPWRS.2019.2941029.

[25] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, Advances in Neural Information Processing Systems 30 (2017).

[26] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, Y. Bengio, Graph attention networks, in: International Conference on Learning Representations, 2018.

[27] P. M. Anderson, A.-A. A. Fouad, Power System Control and Stability, 2nd Edition, Wiley-IEEE Press, Piscataway, NJ, 2003.

[28] J. Machowski, Z. Lubosny, J. W. Bialek, J. R. Bumby, Power System Dynamics: Stability and Control, 3rd Edition, John Wiley & Sons, Chichester, UK, 2020.

[29] P. Kundur, J. Paserba, V. Ajjarapu, G. Andersson, A. Bose, C. Canizares, N. Hatziargyriou, D. Hill, A. Stankovic, C. Taylor, T. Van Cutsem, V. Vittal, Definition and classification of power system stability IEEE/CIGRE joint task force on stability terms and definitions, IEEE Transactions on Power Systems 19 (2) (2004) 1387–1401.

[30] P. W. Sauer, M. A. Pai, Power System Dynamics and Stability, Prentice Hall, Upper Saddle River, NJ, 1998.

[31] R. D. Zimmerman, C. E. Murillo-Sánchez, R. J. Thomas, MATPOWER: Steady-state operations, planning, and analysis tools for power systems research and education, IEEE Transactions on Power Systems 26 (1) (2011) 12–19.

[32] W. Wang, Y. Xu, M. Khanna, A survey on the communication architectures in smart grid, Computer Networks 55 (15) (2011) 3604–3629.

[33] K. Yu, L. Zhu, Z. Wen, A. Mohammad, Z. Zhou, T. Sato, CCN-AMI: Performance evaluation of content-centric networking approach for advanced metering infrastructure in smart grid, in: IEEE International Workshop on Applied Measurements for Power Systems Proceedings (AMPS), IEEE, 2014, pp. 1–6. doi:10.1109/AMPS.2014.6947719.

[34] North American Electric Reliability Corporation, PRC-005-6: Protection system, automatic reclosing, and sudden pressure relaying maintenance, NERC Reliability Standard (2014).

[35] IEEE Power and Energy Society, IEEE standard for synchrophasor measurements for power systems – IEEE Std C37.118.1-2011, IEEE Std C37.118.1-2011 (2011) 1–61.

[36] F. L. Bauer, C. T. Fike, Norms and exclusion theorems, Numerische Mathematik 2 (1) (1960) 137–141.

[37] G. A. Baker, Jr., P. Graves-Morris, Padé Approximants, 2nd Edition, Cambridge University Press, Cambridge, UK, 1996.

[38] F. Milano, Power System Modelling and Scripting, Springer, London, UK, 2010.

[39] T. M. Cover, J. A. Thomas, Elements of Information Theory, 2nd Edition, John Wiley & Sons, Hoboken, NJ, 2006.

[40] N. Tishby, F. C. Pereira, W. Bialek, The information bottleneck method, in: Proceedings of the 37th Annual Allerton Conference on Communication, Control, and Computing, 2000, pp. 368–377.

[41] P. Shaw, J. Uszkoreit, A. Vaswani, Self-attention with relative position representations, in: Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT), 2018, pp. 464–468.

[42] T. Beucler, M. Pritchard, S. Rasp, J. Ott, P. Baldi, P. Gentine, Enforcing analytic constraints in neural networks emulating physical systems, Physical Review Letters 126 (9) (2021) 098302.

[43] International Electrotechnical Commission, IEC 61850: Communication networks and systems for power utility automation, International Standard, originally published 2003, updated 2020 (2020).

[44] F. Wilcoxon, Individual comparisons by ranking methods, Biometrics Bulletin 1 (6) (1945) 80–83.

[45] S. Holm, A simple sequentially rejective multiple test procedure, Scandinavian Journal of Statistics 6 (2) (1979) 65–70.

[46] J. Demšar, Statistical comparisons of classifiers over multiple data sets, Journal of Machine Learning Research 7 (2006) 1–30.